

Version 5.70

Customer : V I N H S O N
Plant Designation : TURBINE CONTROL CUBICLE
Drawing No. : VIN12-GOV-0021
Commission : CIRCUIT DIAGRAM

Manufacturer : ANDRITZ HYDRO
Lunzerstr. 78
4031 LINZ
A U S T R I A

Cubicle Manu. : RITTAL
Typ : TS 8806.521
Loc. of install. :
Special feature :
Special remarks : RAL 7035
Location :
Regulationes : ÖVE EN1 / DIN VDE / EN 60439
Degree of protec. : IP 55 (EN 60 529)

Dim. of Cubicle : WxHxD= 800x2000x600mm
+ Plinth H=200mm

Power Supply : 1x 125VDC, 1x 220VAC
Feeding Cable :
Control Voltage : 24VDC
Year of Constr. : 2012

Project Start : 30.04.2012
Project Responsible : FUCHS
Last Revision : 20.07.2012
Letzter Editor : AUER

Highest Sheet No. : 69
Number of Sheets : 69

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	01 Workshop As Built	07-2012	Fuchs	Des./Appr.				30.04.2012	Fuchs	Project-No.	Zeich./Draw.No.	Index
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7 = Zeilenabstand in mm

1 = Spaltenanzahl

1 = Spaltenabstand

1 = Richtung (senkrecht)

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01 Workshop As Built

Änderung / Modification

07-2012

Datum/Date

Fuchs

Bearb. /Des.

30.04.2012

Wassstab/Scale

Fuchs

30.04.2012

Last Edit:

20.07.2012

ANDRITZ

Hydro

TURBINE CONTROL CUBICLE

Circuit Diagramm

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=VINHSON

+MEY10GH100

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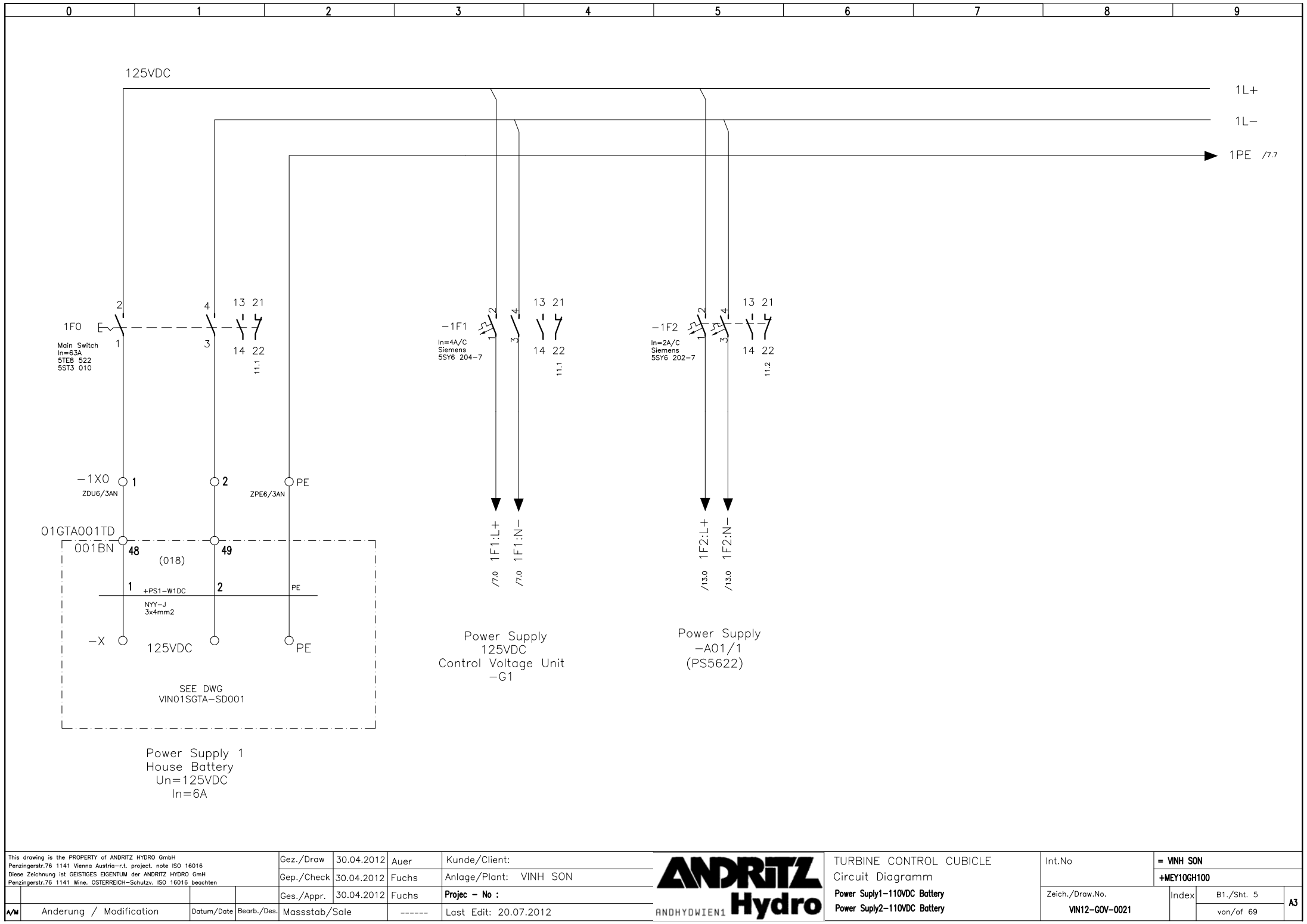
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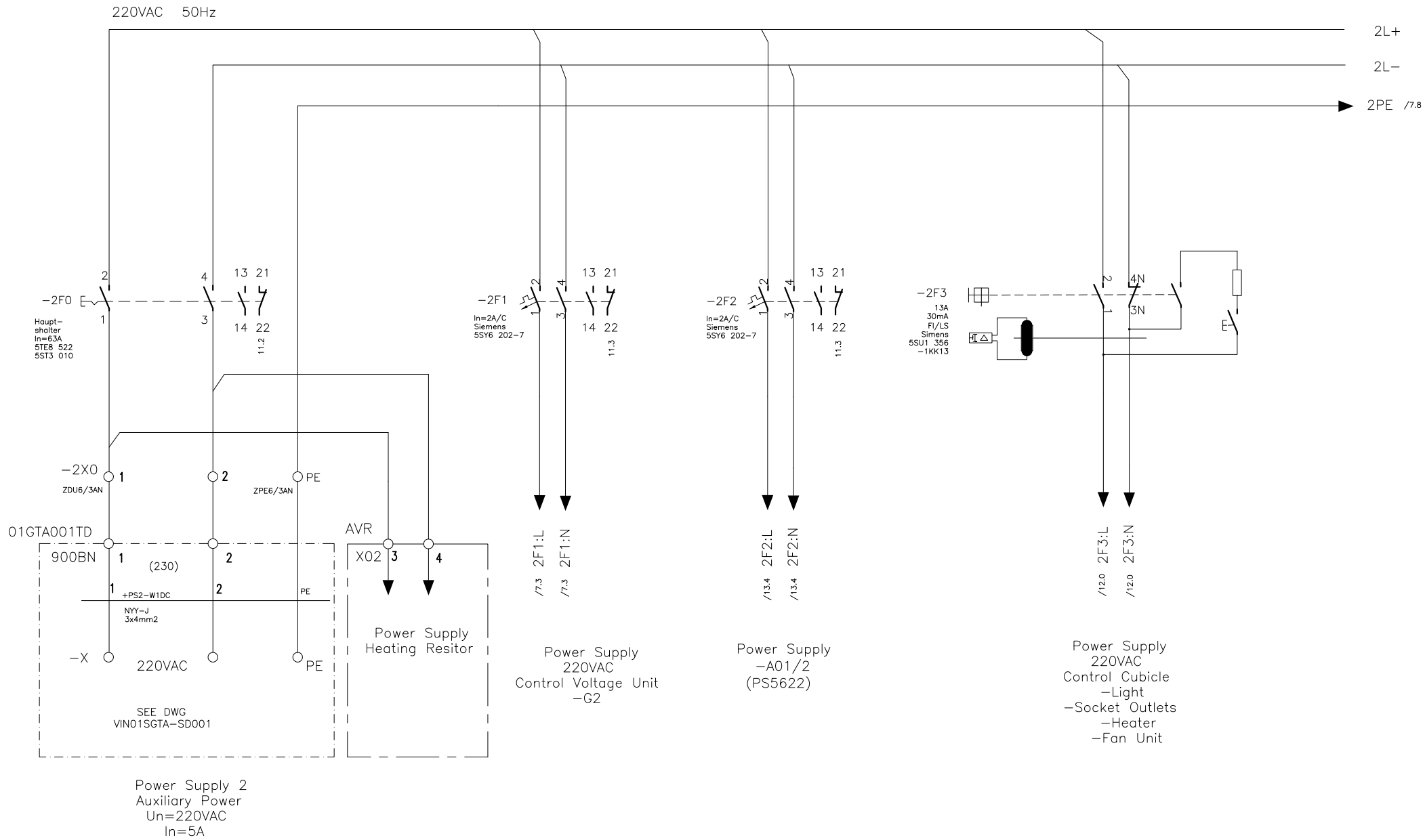
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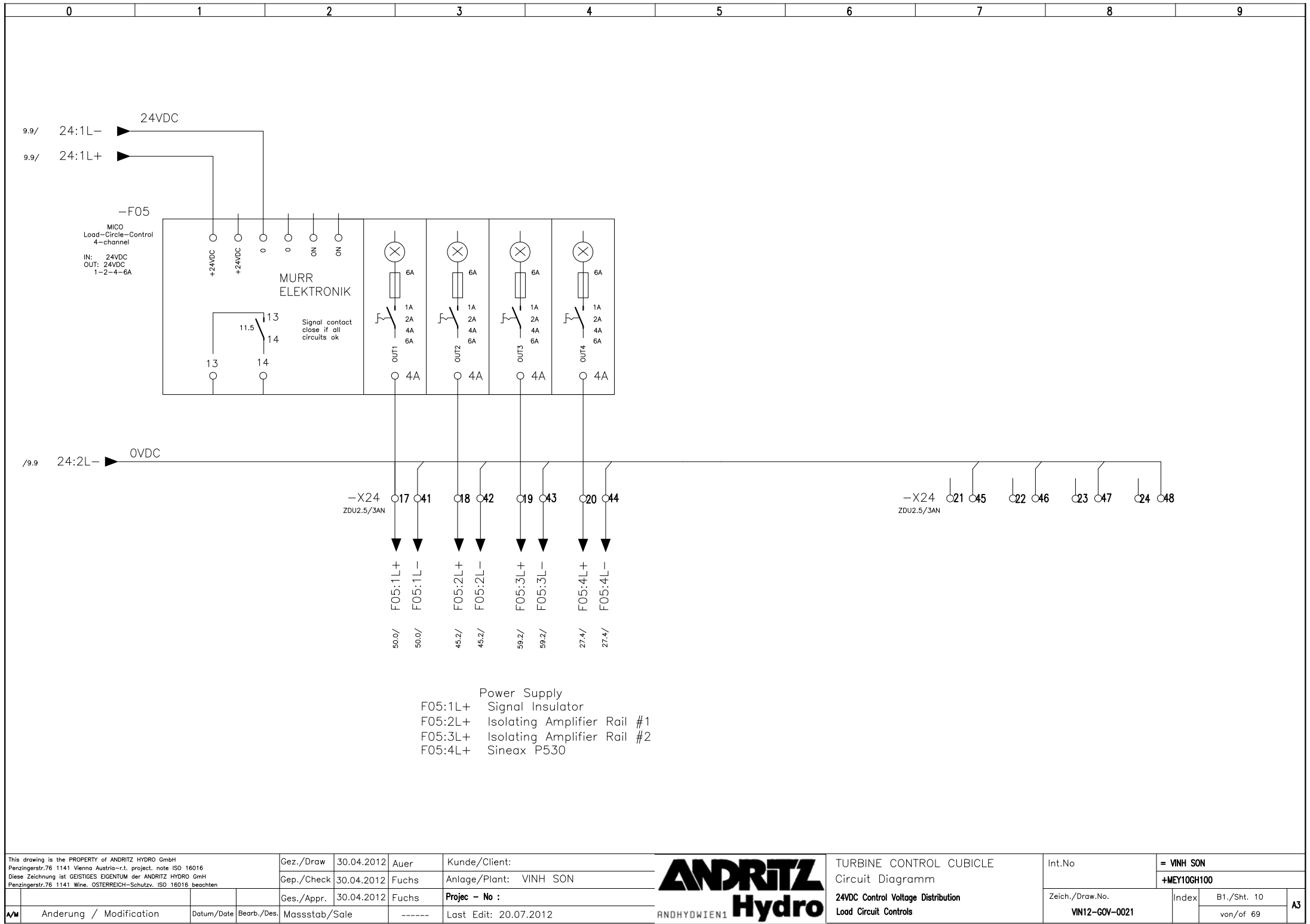
INH 01.2007

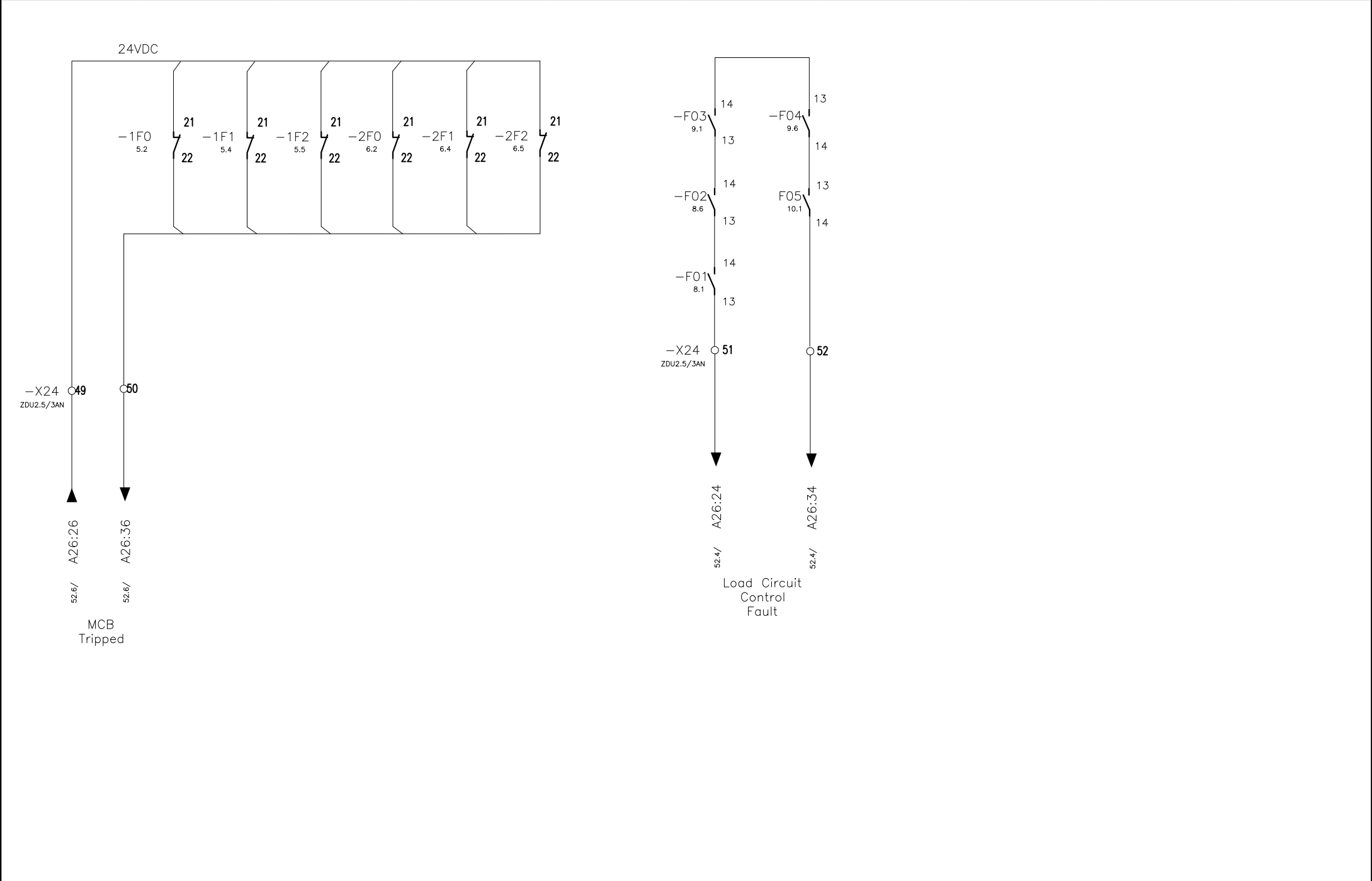
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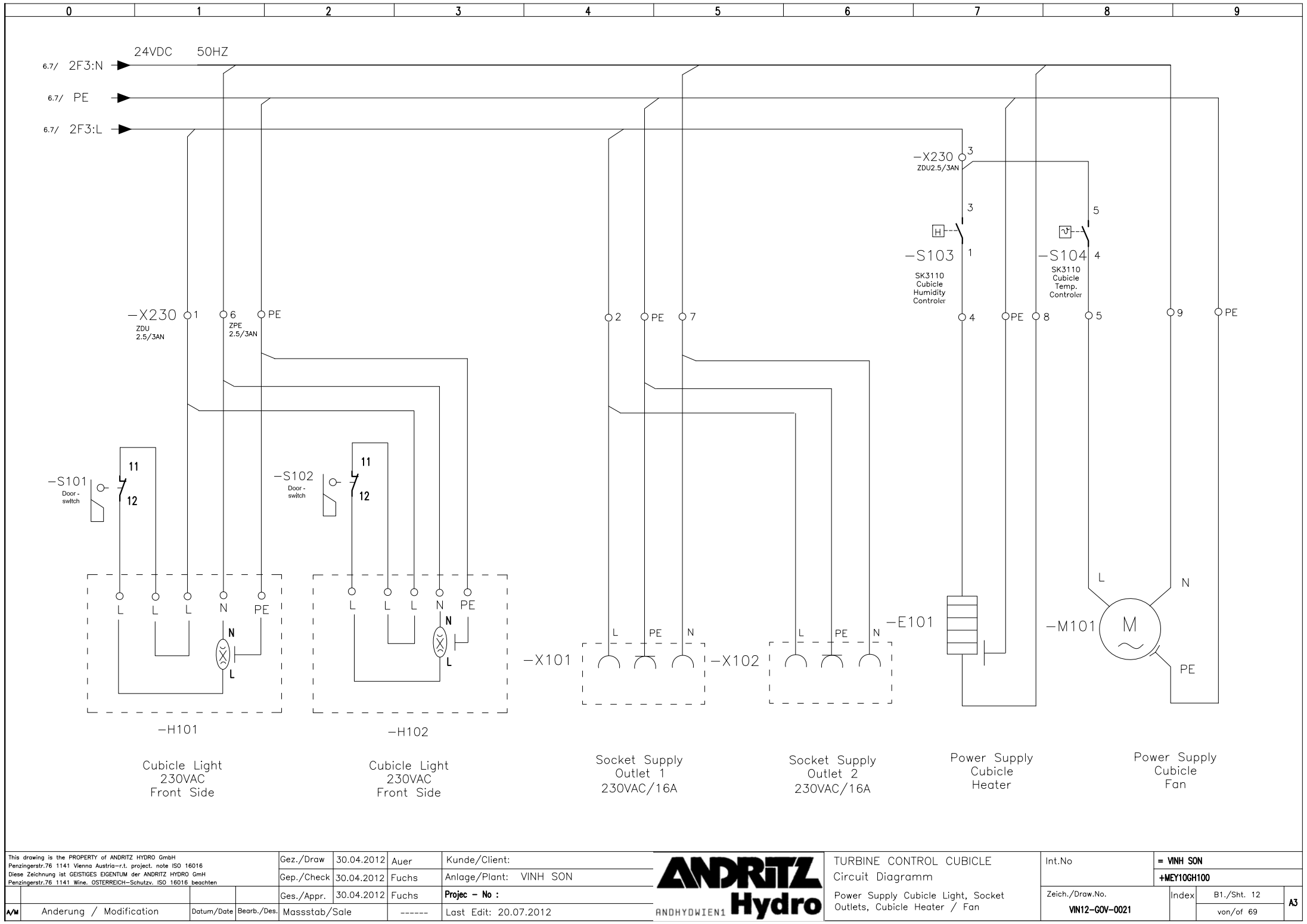
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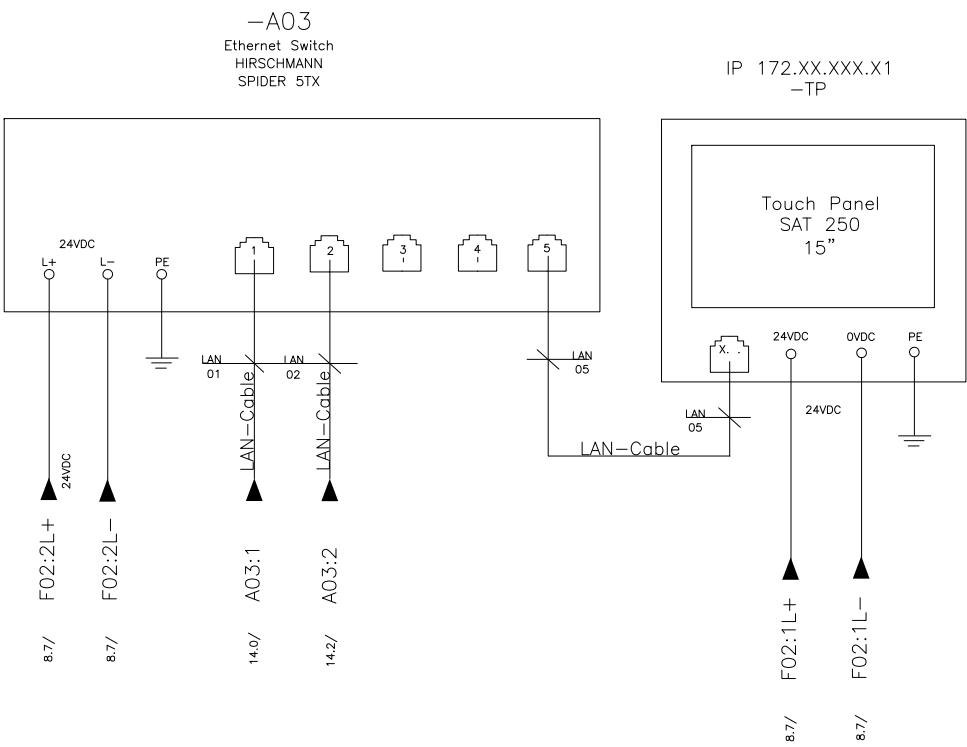












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				Ges./Appr.	30.04.2012	Fuchs	Projec - No :
				Masstab/Sale	-----		Last Edit: 11.05.2012

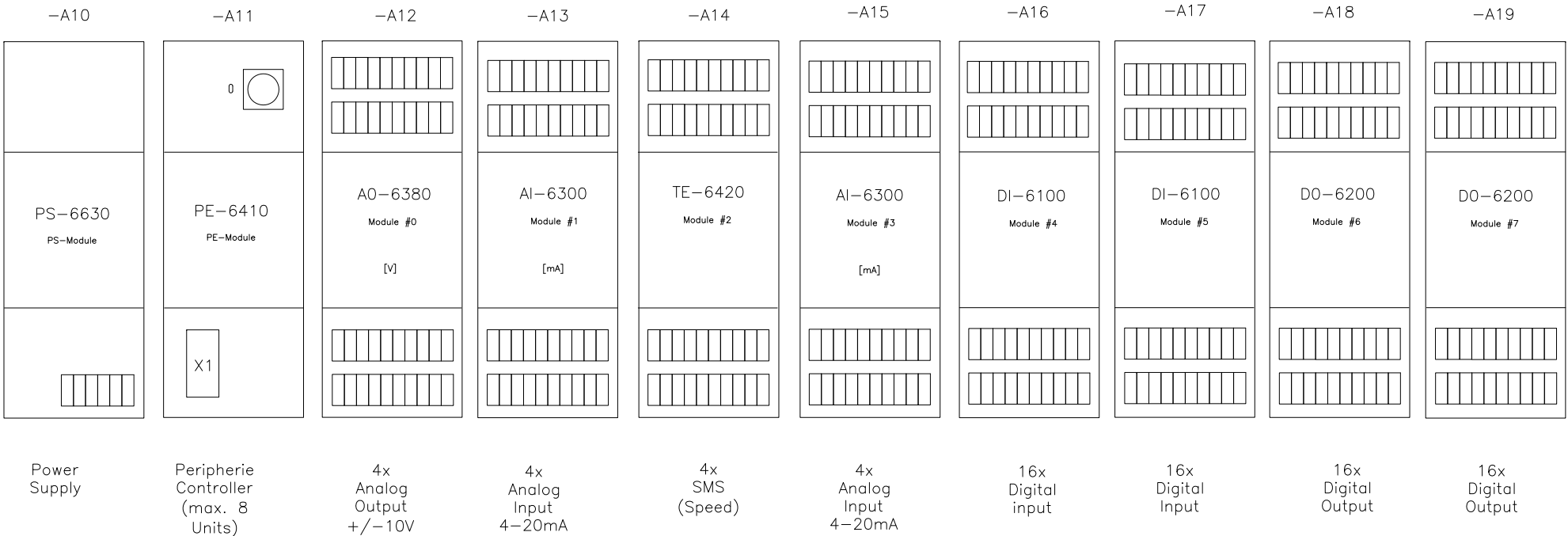


TURBINE CONTROL CUBICLE
Circuit Diagramm
Circuit Diagramm
Ethernet Switch, Touch Panel

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	+MEY10GH100		
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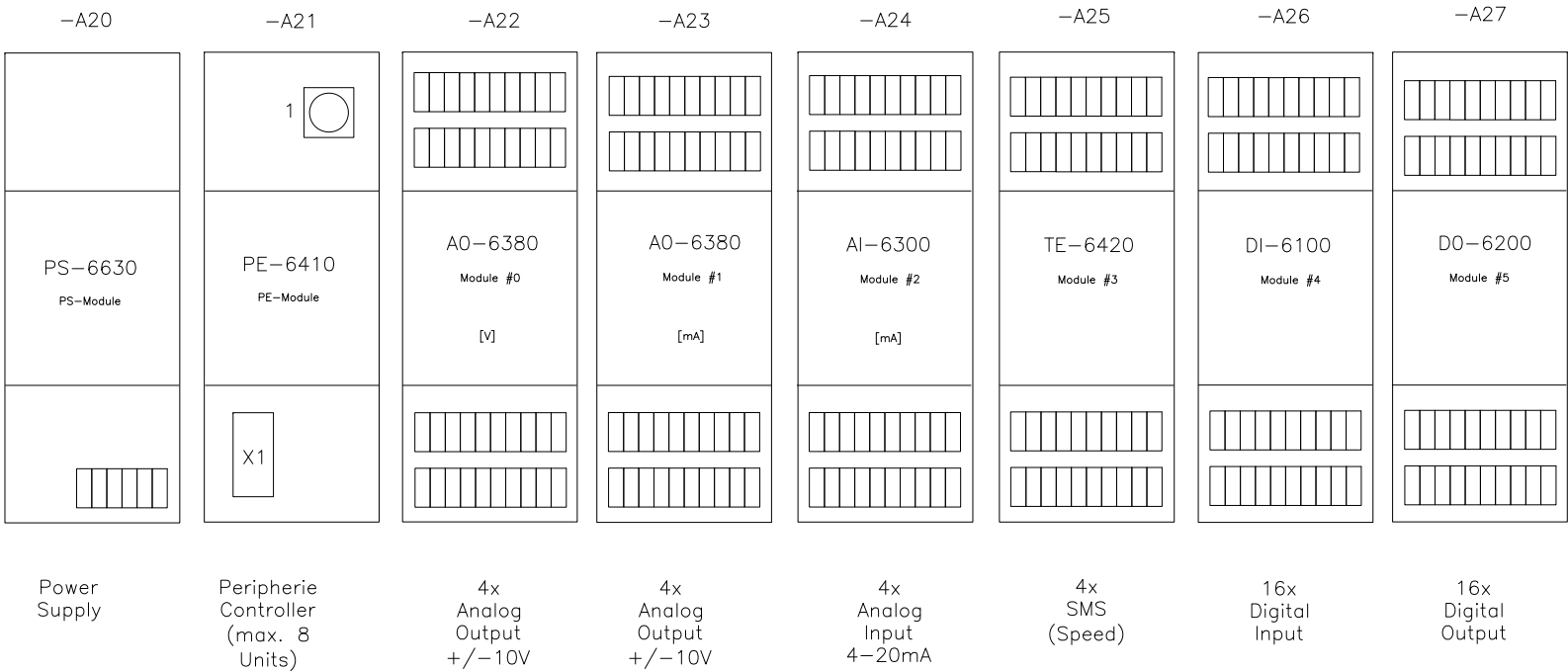
Terminal Module

Rail #0



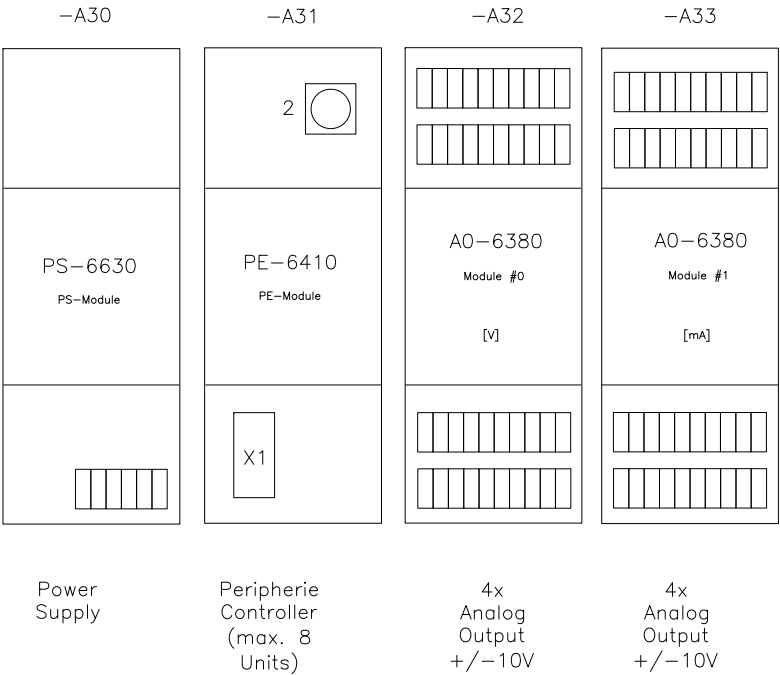
Terminal Module

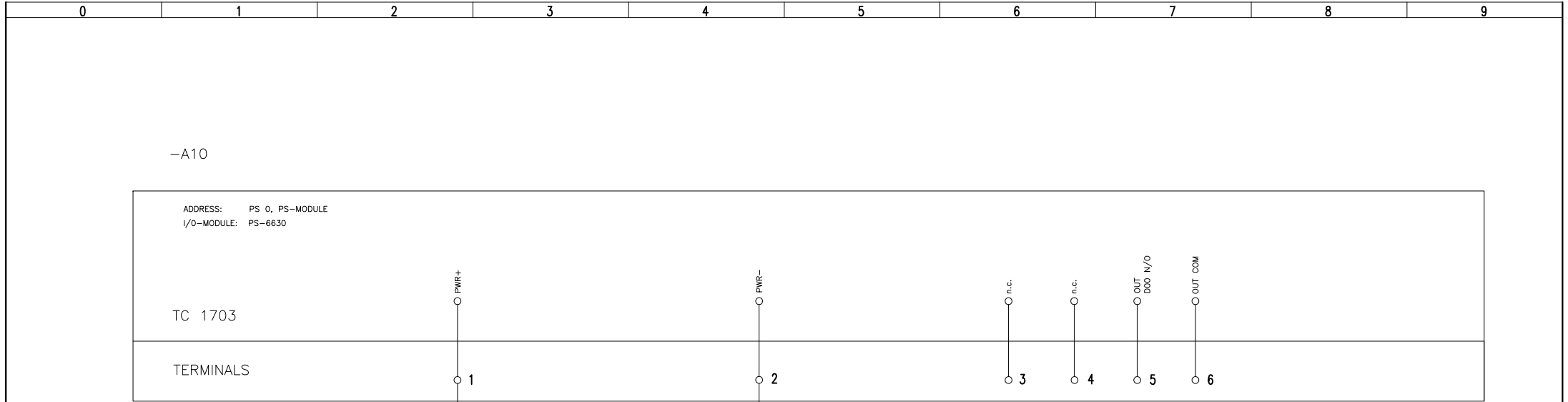
Rail #1



Terminal Module

Rail #2





0	1	2	3	4	5	6	7	8	9
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—A11

ADDRESS: PE-MODULE
MODULE: PE-6401

TC 1703

X1

14.7/ USB/1

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				Gep./Check 30.04.2012 Fuchs Anlage/Plant: VINH SON						
				Ges./Appr. 30.04.2012 Fuchs Projec - No :			Peripherycontroller #1			
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AO 2
+/-10V
Control Signal Needle 1
-100%...+100%
SIG-N1

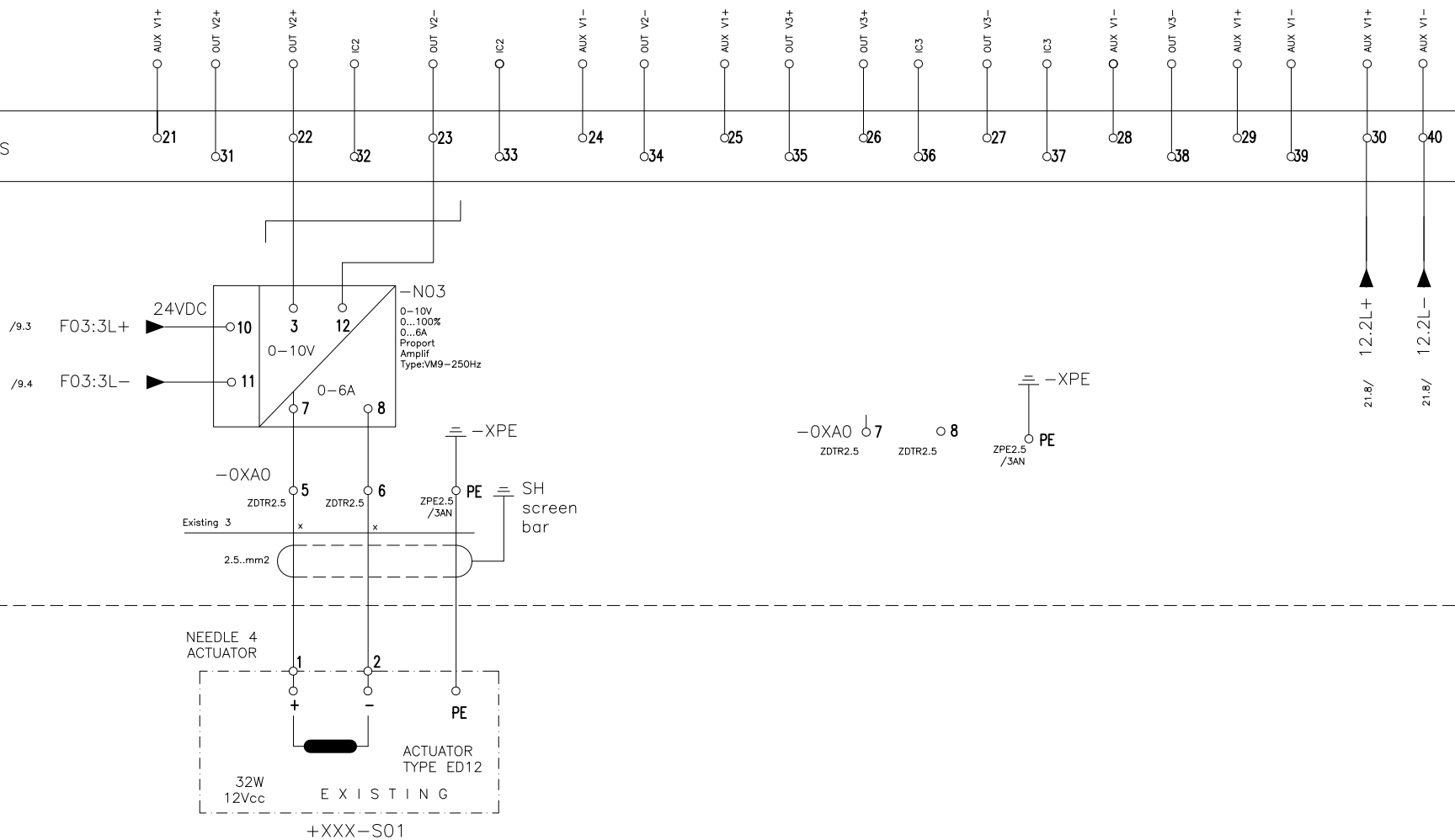
A0 3
+/-10V
S P A R E

-A12

ADDRESS: PE 0, MODULE 0
I/O-MODULE: A0-6380

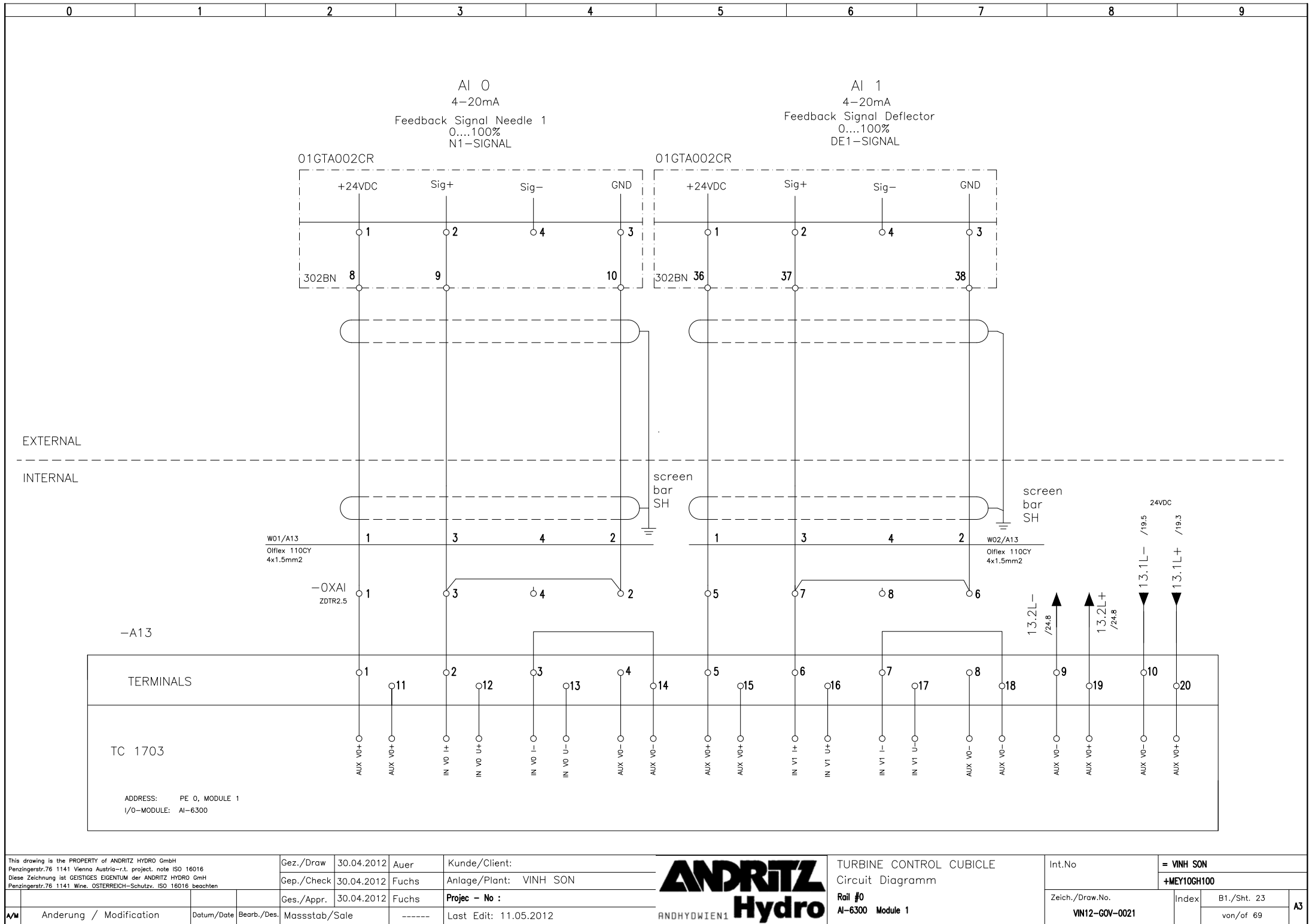
TC 1703

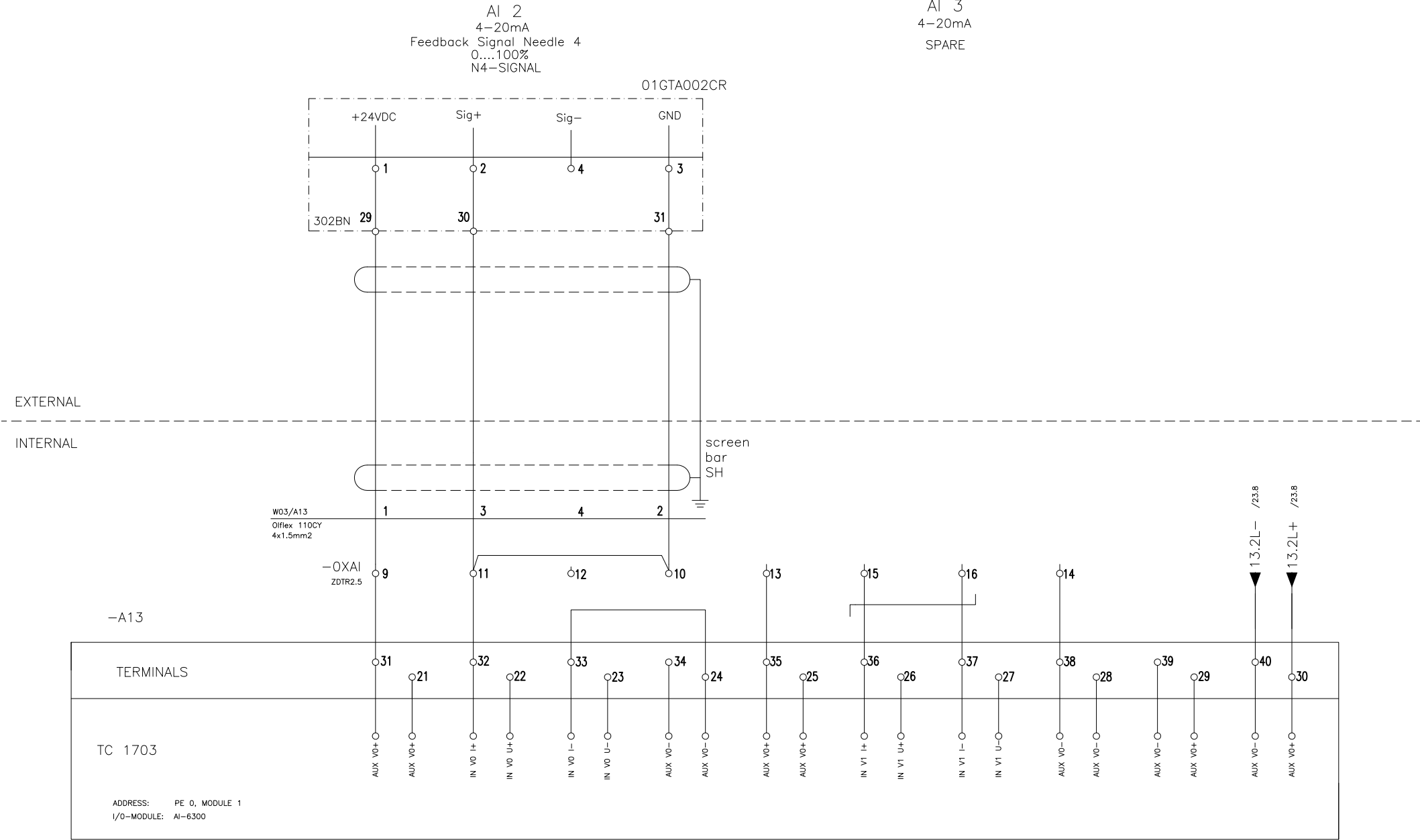
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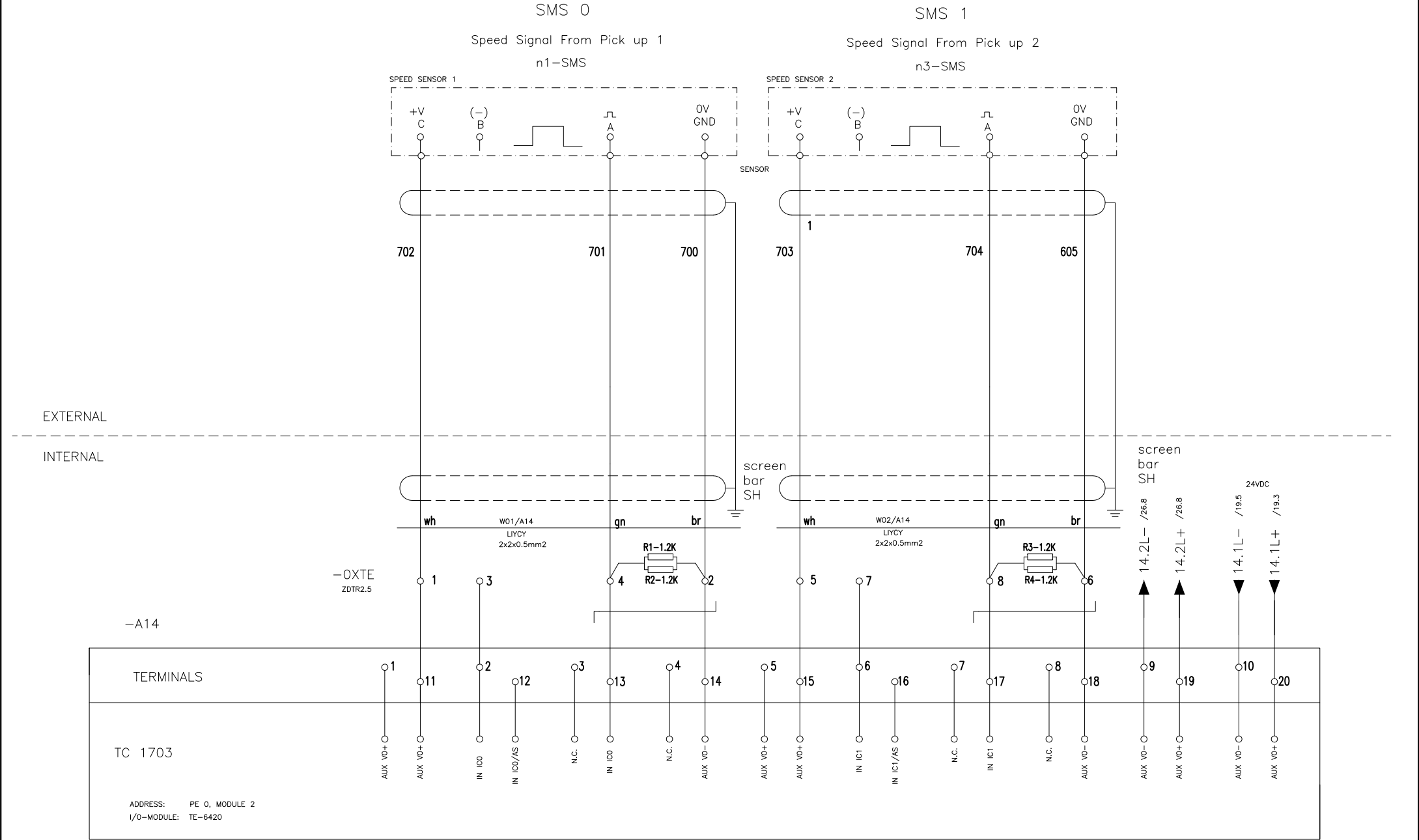


INTERNAL

EXTERNAL







SMS 3

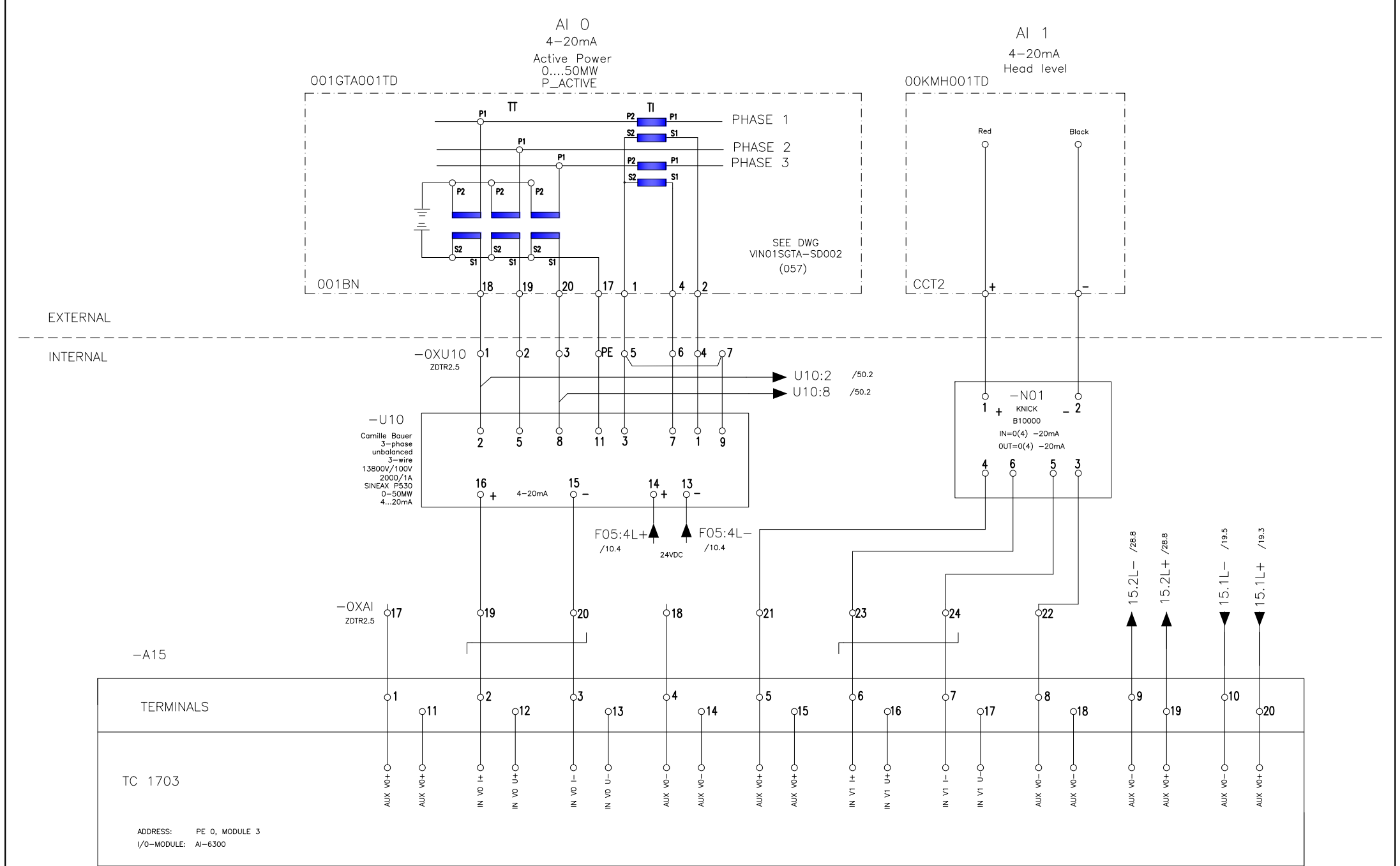
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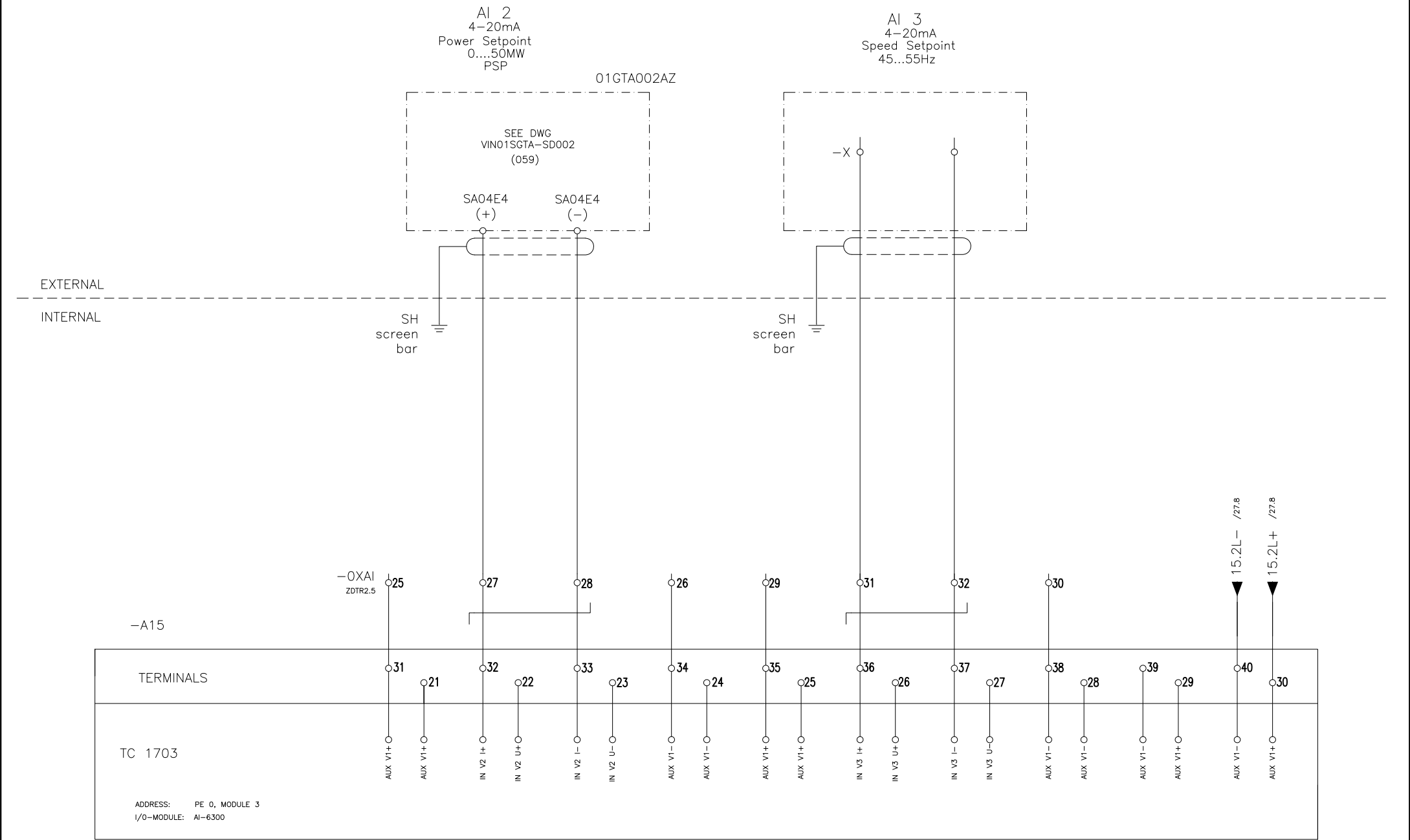
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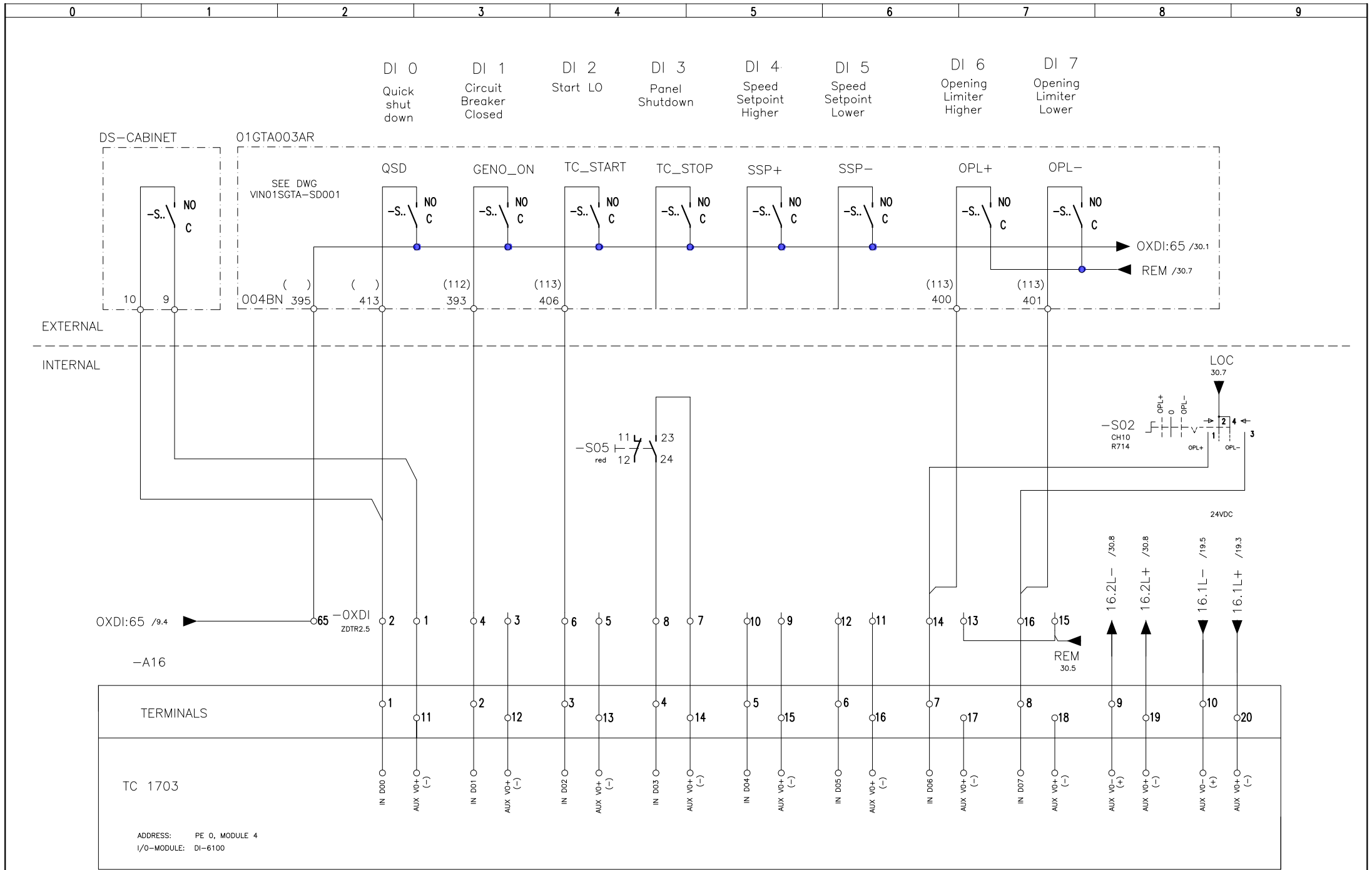


TC 1703

ADDRESS: PE 0, MODULE 2
I/O-MODULE: TE-6420







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Gep./Check 30.04.2012 Fuchs Anlage/Plant: VINH SON

Ges./Appr. 30.04.2012 Fuchs Projec - No :

Last Edit: 11.05.2012



TURBINE CONTROL CUBICLE
Circuit Diagramm
Rail #0
DI-6100 Module 4

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= VINH SON

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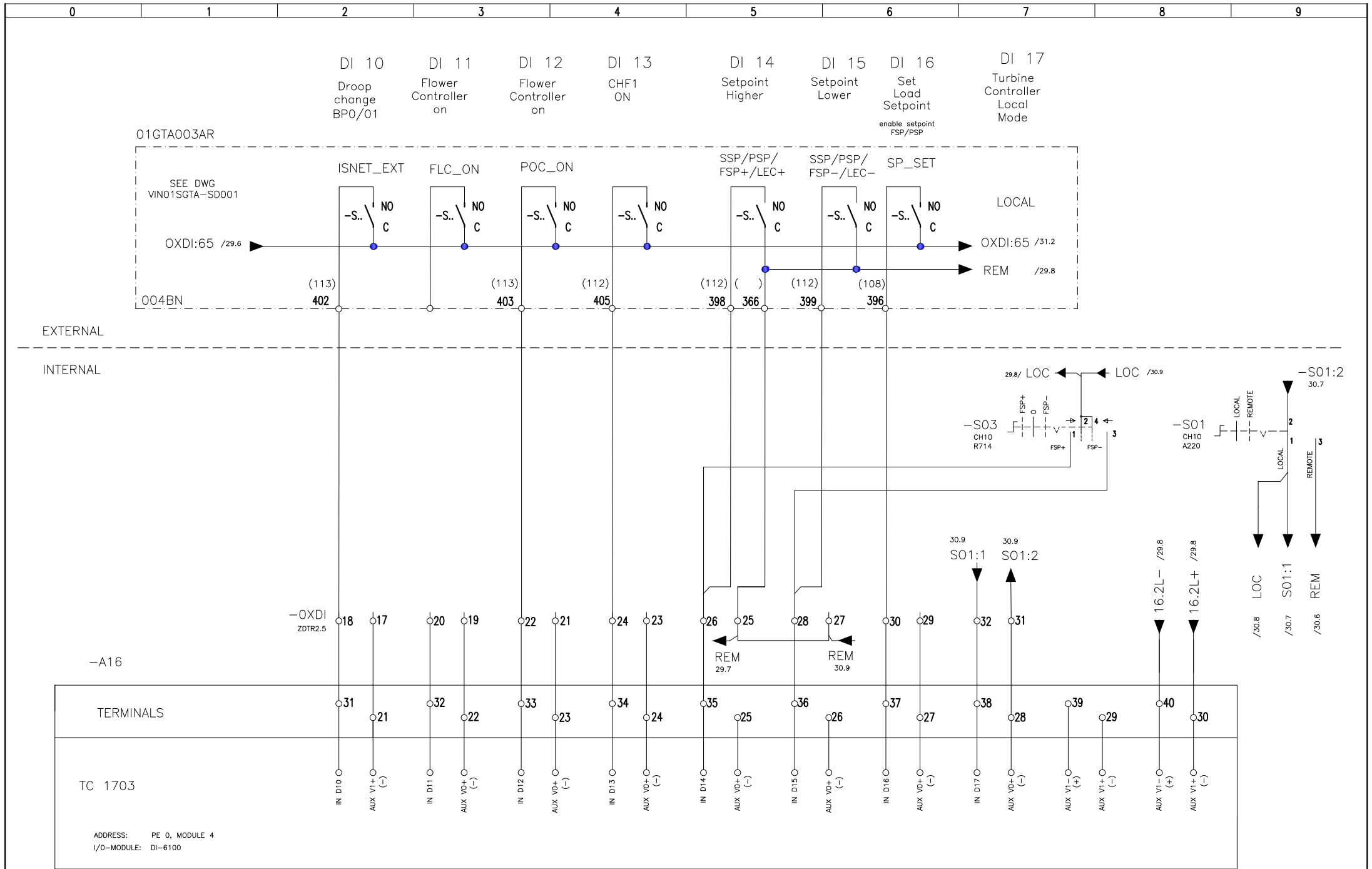
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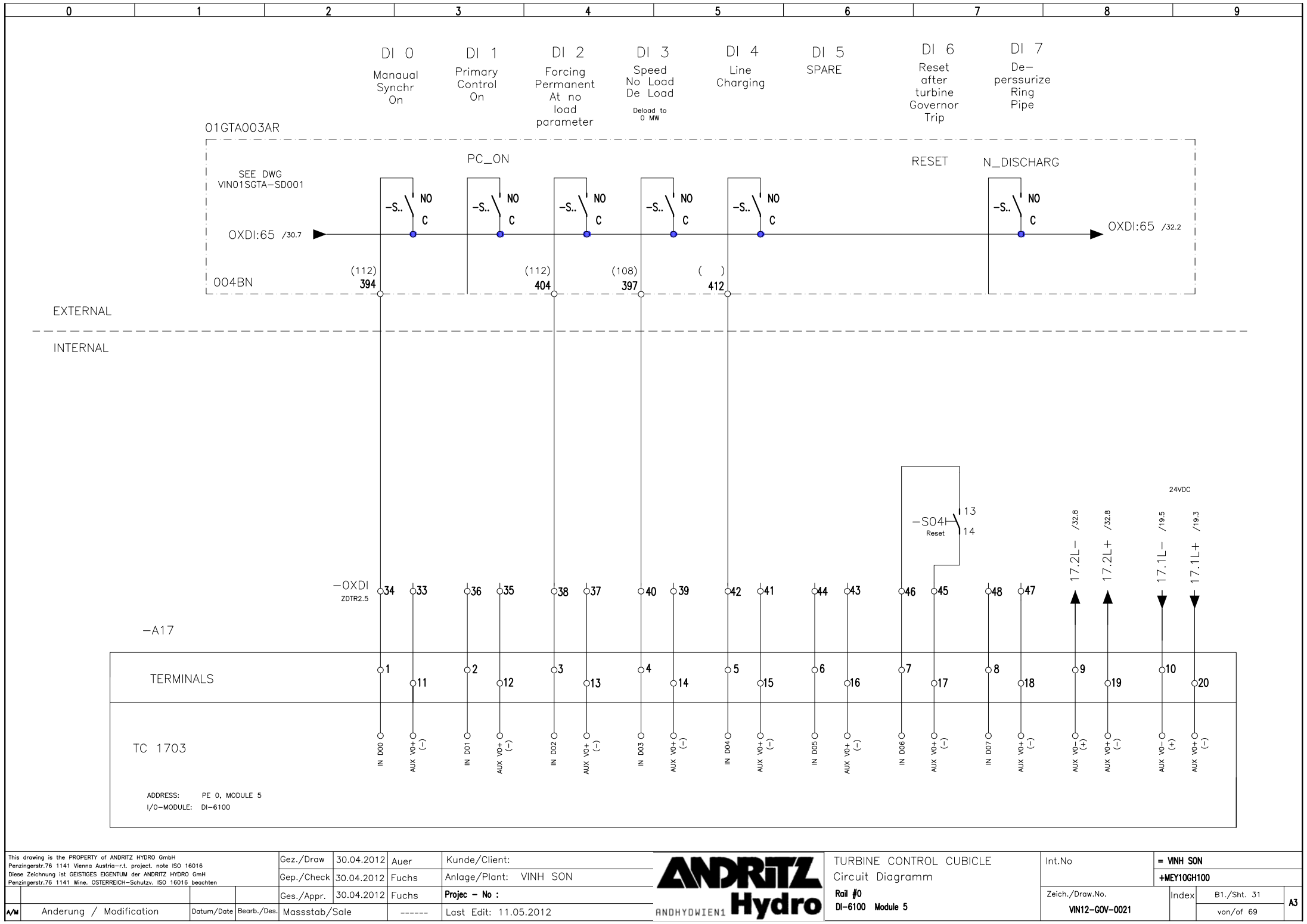
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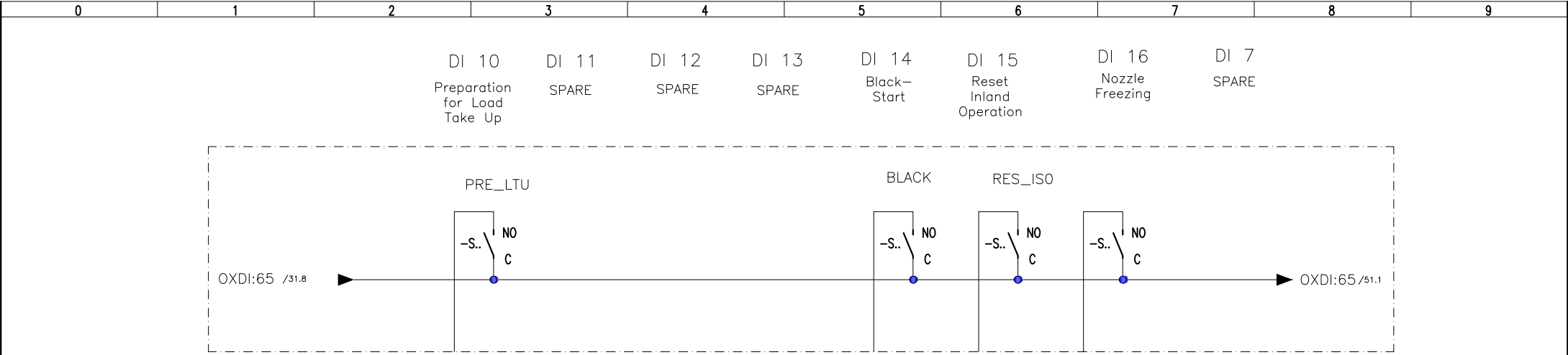
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Bearb./Des.

Massstab/Sale



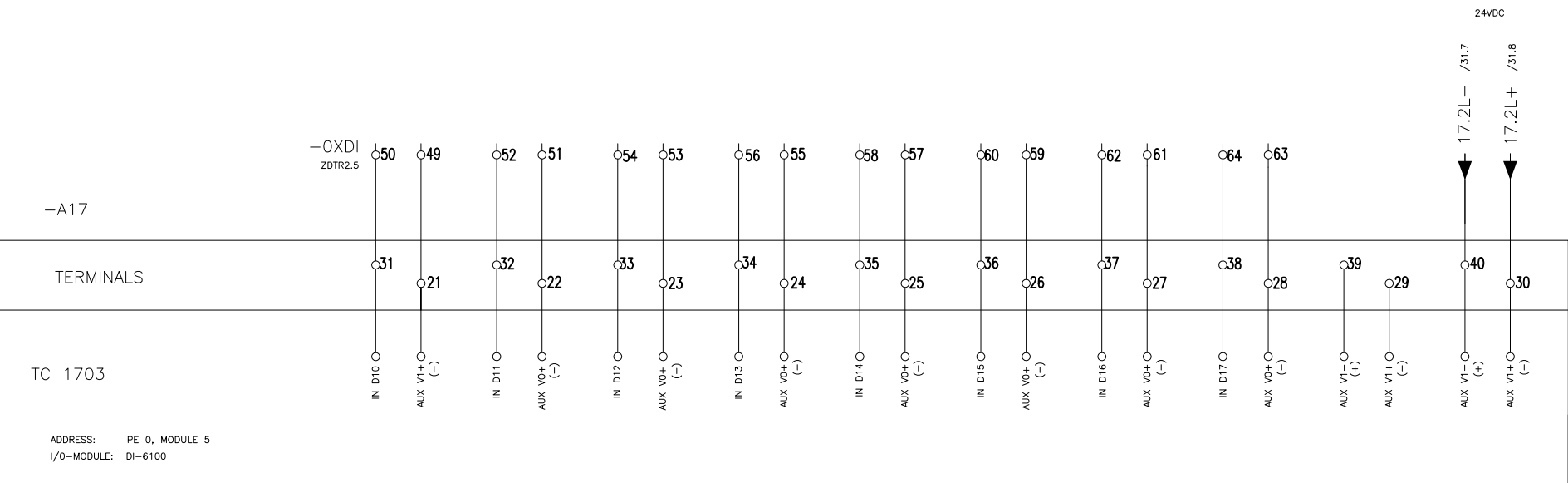




EXTERNAL

INTERNAL

-A17



ADDRESS: PE 0, MODULE 5
I/O-MODULE: DI-6100

-A18	D0 0 Major Fault No Turbine Governor Shut Down NOT-ESD-QSD	D0 1 Minor Fault General Alarm FAIL-IT	D0 2 Turbine Limited TU-LIMITED	D0 3 Isolating Operation Active ACTIVE	D0 4 Forcing parameter at no load
ADDRESS: PE 0, MODULE 6 I/O-MODULE: DO-6200					
TC 1703	OUT D00	OUT D01	OUT D02	OUT D03	OUT D04
	AUX V0-	AUX V0-	AUX V0-	AUX V0-	AUX V0-
TERMINALS	11	12	13	14	15
	1	2	3	4	5
-K00 24VDC -K01 24VDC -K02 24VDC -K03 24VDC -K04 24VDC					
EXTERNAL					
	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24
	14 11 63.1 12 21 63.2 24 31 22 41 34 41 32 41 44 41 42 41	14 11 63.4 12 21 63.4 24 21 22 31 34 31 32 41 44 41 42 41	14 11 63.5 12 21 63.5 24 21 22 31 34 31 32 41 44 41 42 41	14 11 63.6 12 21 63.6 24 21 22 31 34 31 32 41 44 41 42 41	14 11 63.8 12 21 63.8 24 21 22 31 34 31 32 41 44 41 42 41

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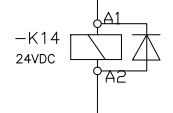
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DO 14
Speed Switch
Excitation
n<70%

TERMINALS

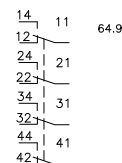
— V1 AUX —

35



EXTERNAL

Schrack
PT570L24



A3

D0 17
Speed Switch
SPARE

TERMINALS

—AUX V1—

DAILY V1 =

1

1

1821

EXTERNAL

Schrack
PT570L24

64.7

	D0 0 Needle 1 not bloked	D0 1 Needle 2 not bloked	D0 2 Needle 3 not bloked	D0 3 Needle 4 not bloked	D0 4 Deflector not bloked
-A19					
ADDRESS: PE 0, MODULE 7 I/O-MODULE: D0-6200					
TC 1703					
TERMINALS	OUT D00 VO-	OUT D01 VO-	OUT D02 VO-	OUT D03 VO-	OUT D04 VO-
	AUX VO-	AUX VO-	AUX VO-	AUX VO-	AUX VO-
	11	12	13	14	15
	1	2	3	4	5
INTERNAL					
EXTERNAL					
	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24	Schrack PT570L24

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				Gep./Check	30.04.2012	Fuchs	Anlage/Plant: VINH SON
				Ges./Appr.	30.04.2012	Fuchs	Projec - No :
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Rail #0 D0-6200 Module 7							
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Anderung / Modification									

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Ges./Appr. 30.04.2012 Fuchs Projec - No :

ANDRITZ
Hydro
ANDHYDOWIEN1

TURBINE CONTROL CUBICLE
Circuit Diagramm
Rail #0
D0-6200 Module 7

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INTERNAL

EXTERNAL

TC 1703

TERMINALS

ADDRESS: PE 0, MODULE 7
I/O-MODULE: D0-6200

D0 10 Ready for Load Take Up R_LTU
D0 11 Unit in Operation U_RUN
D0 12 Black Start Active
D0 13 Speed Signal Faulty
D0 14 Speed Noload position

-A19

OUT D10 AUX V1-
OUT D11 AUX V1-
OUT D12 AUX V1-
OUT D13 AUX V1-
OUT D14 AUX V1-

21 31 22 32 23 33 24 34 25 35

-K30 24VDC
-K31 24VDC
-K32 24VDC
-K33 24VDC
-K34 24VDC

A1 A2

Schrack PT570L24

14 11 66.1
12 24 21 22 34 31 32 44 41 42

Schrack PT570L24

14 11 66.3
12 24 21 22 34 31 32 44 41 42

Schrack PT570L24

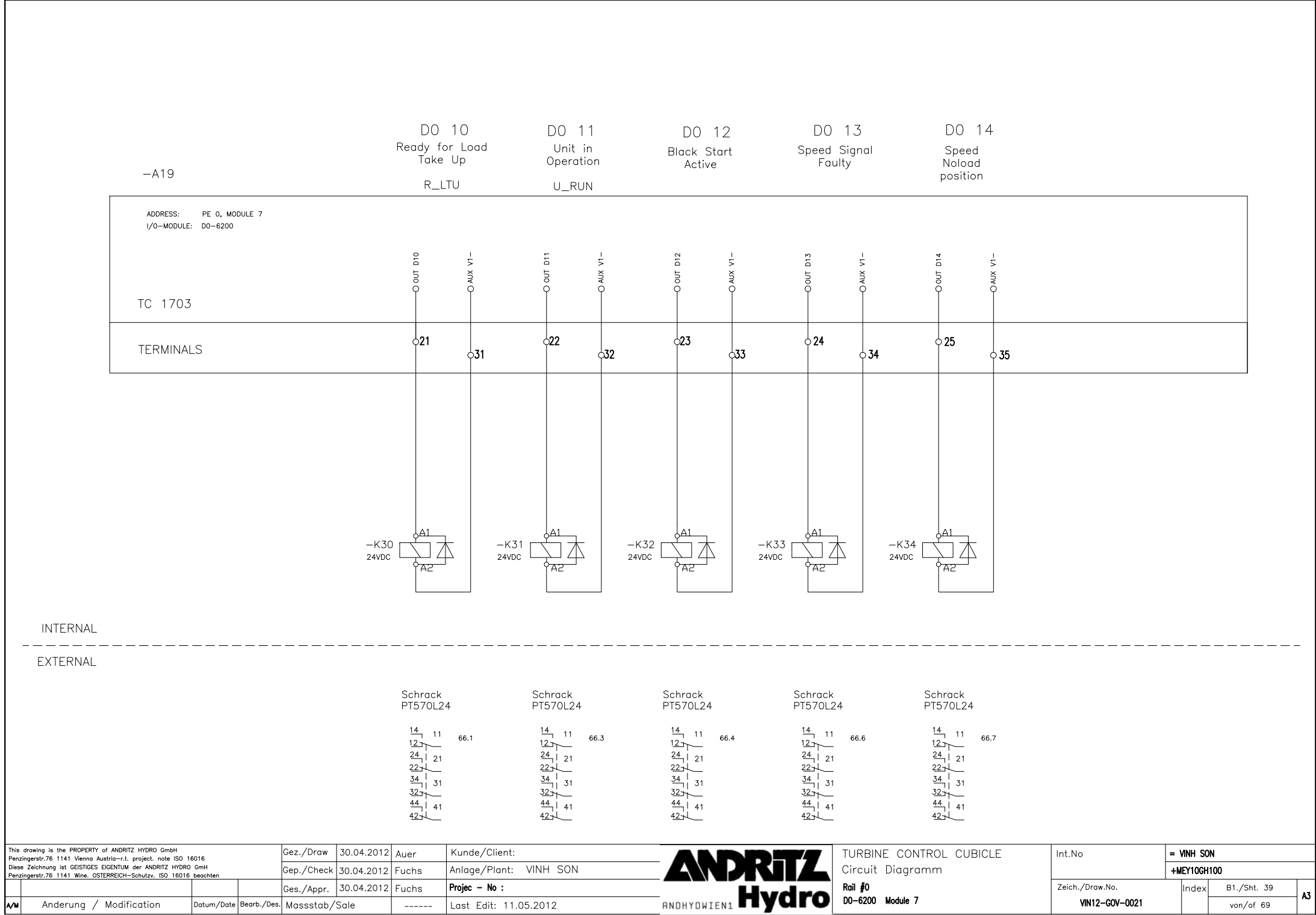
14 11 66.4
12 24 21 22 34 31 32 44 41 42

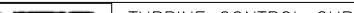
Schrack PT570L24

14 11 66.6
12 24 21 22 34 31 32 44 41 42

Schrack PT570L24

14 11 66.7
12 24 21 22 34 31 32 44 41 42



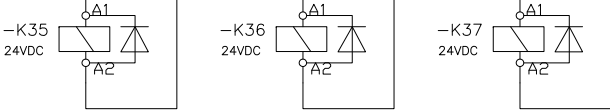
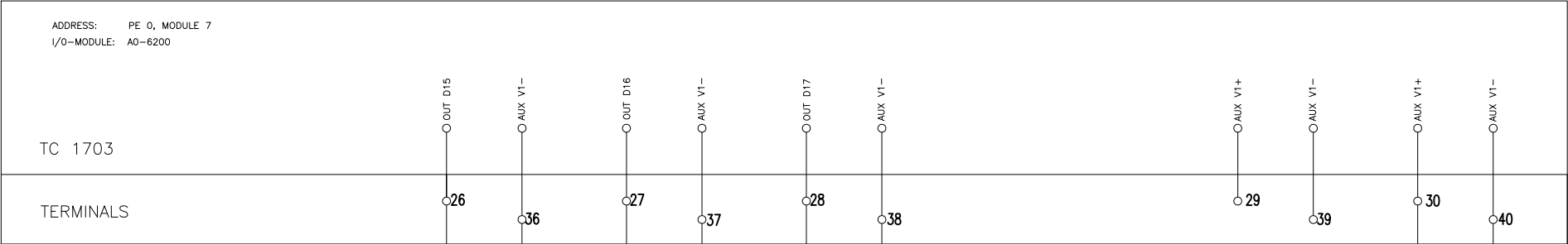
This drawing is the PROPERTY of ANDRITZ HYDRO GmbH Penzingerstr.76 1141 Vienna Austria-r.t. project. note ISO 16016 Diese Zeichnung ist GEISTIGES EIGENTUM der ANDRITZ HYDRO GmbH Penzingerstr.76 1141 WIne. ÖSTERREICH-Schutz. ISO 16016 beachten				Gez./Draw	30.04.2012	Auer	Kunde/Client:	 ANDHYDWIEN1	TURBINE CONTROL CUBICLE Circuit Diagramm	Int.No	= VINH SON +MEY10GH100		
				Gep./Check	30.04.2012	Fuchs	Anlage/Plant: VINH SON						
				Ges./Appr.	30.04.2012	Fuchs	Projec - No :						
Änderung / Modification				Datum/Date	Bearb./Des.	Massstab/Sale	-----	Last Edit: 11.05.2012	Rail #0 D0-6200 Module 7	Zeich./Draw.No. VIN12-GOV-0021	Index	B1./Sht. 39 von/of 69	A3

D0 15
Start
LO_ON
feedback

D0 16
Primary Control
Active
PC_ON

D0 17
Nozzles
Are Closed
N=0%

–A19

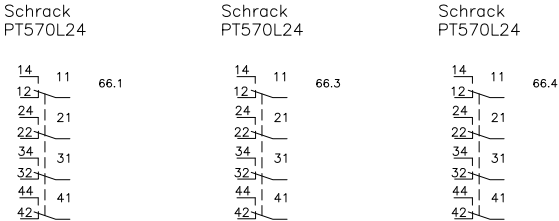


19.2L +
/38.7

19.2L -
/38.7

INTERNAL

EXTERNAL

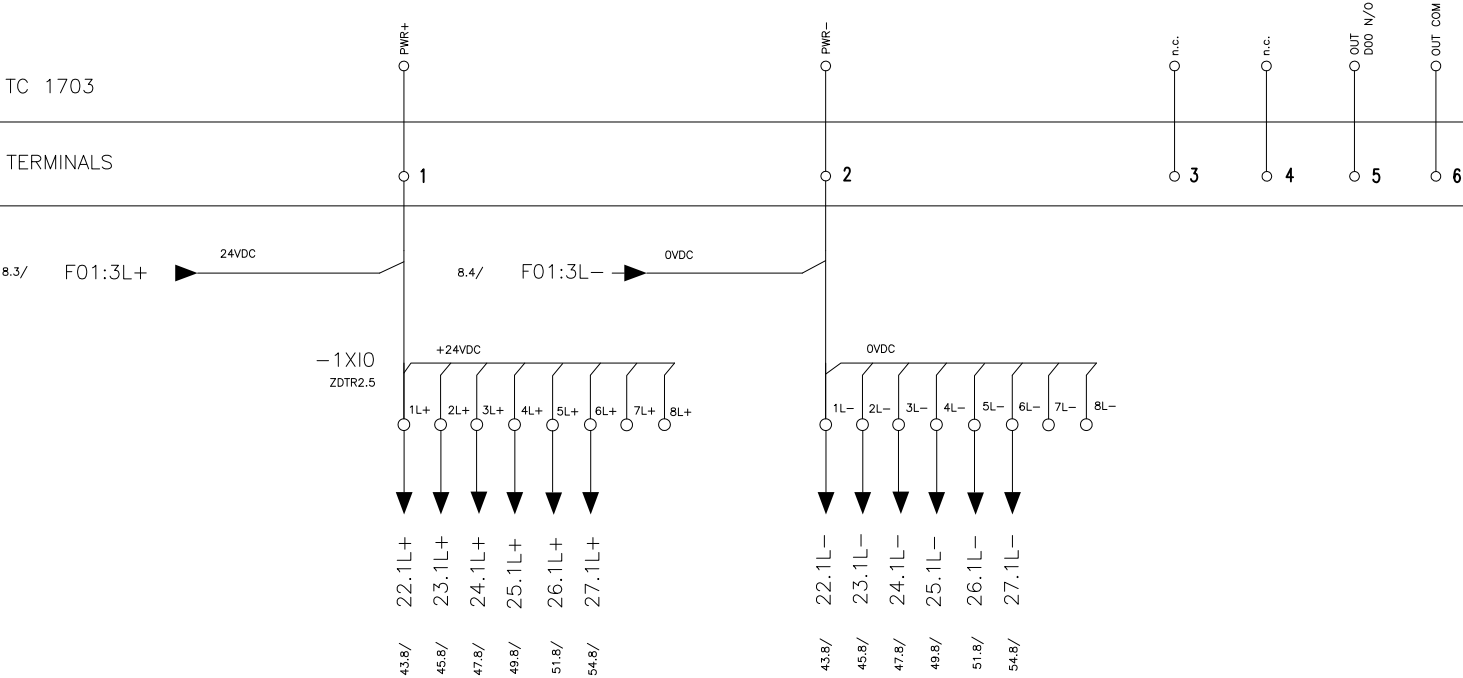


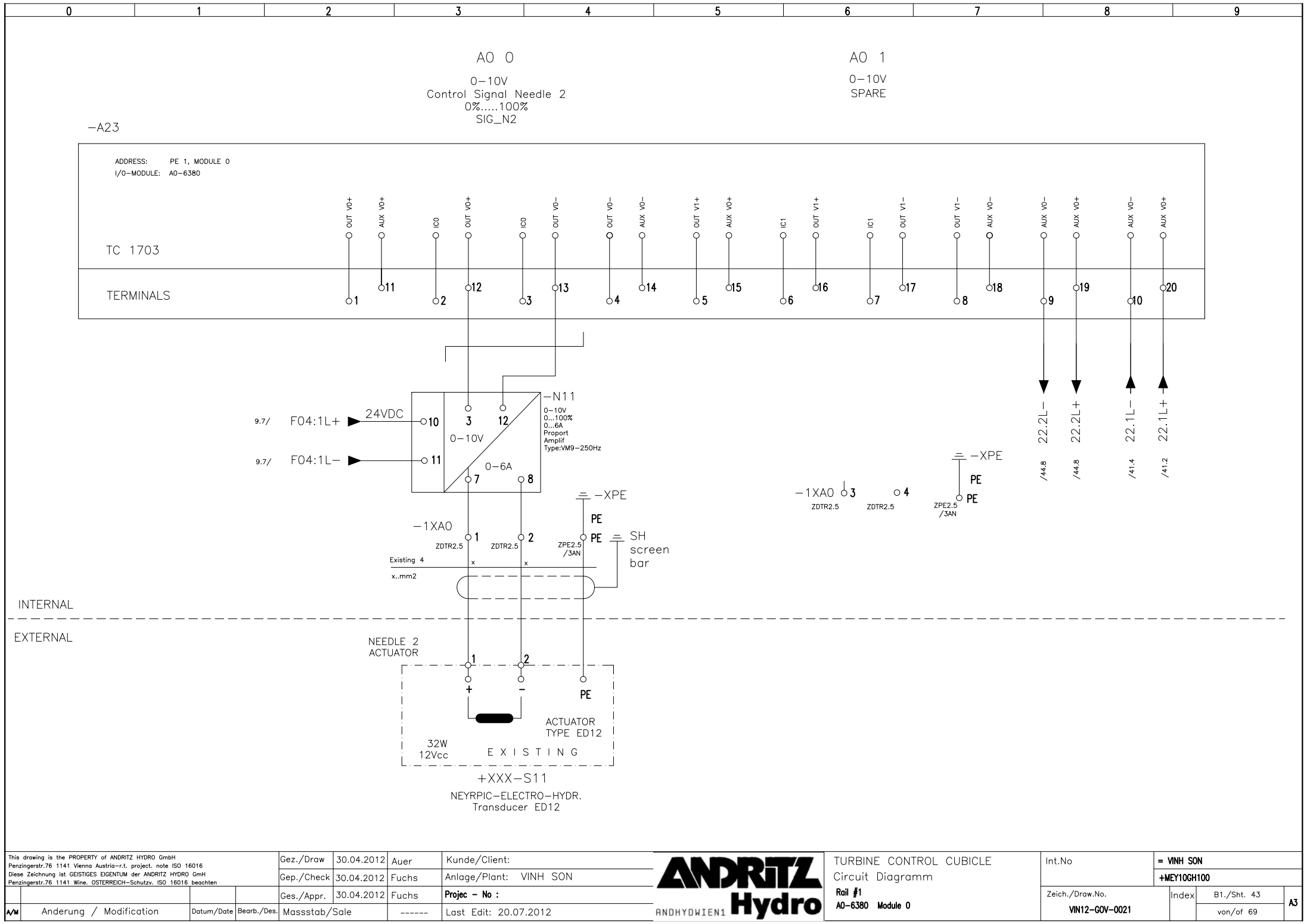
–A20

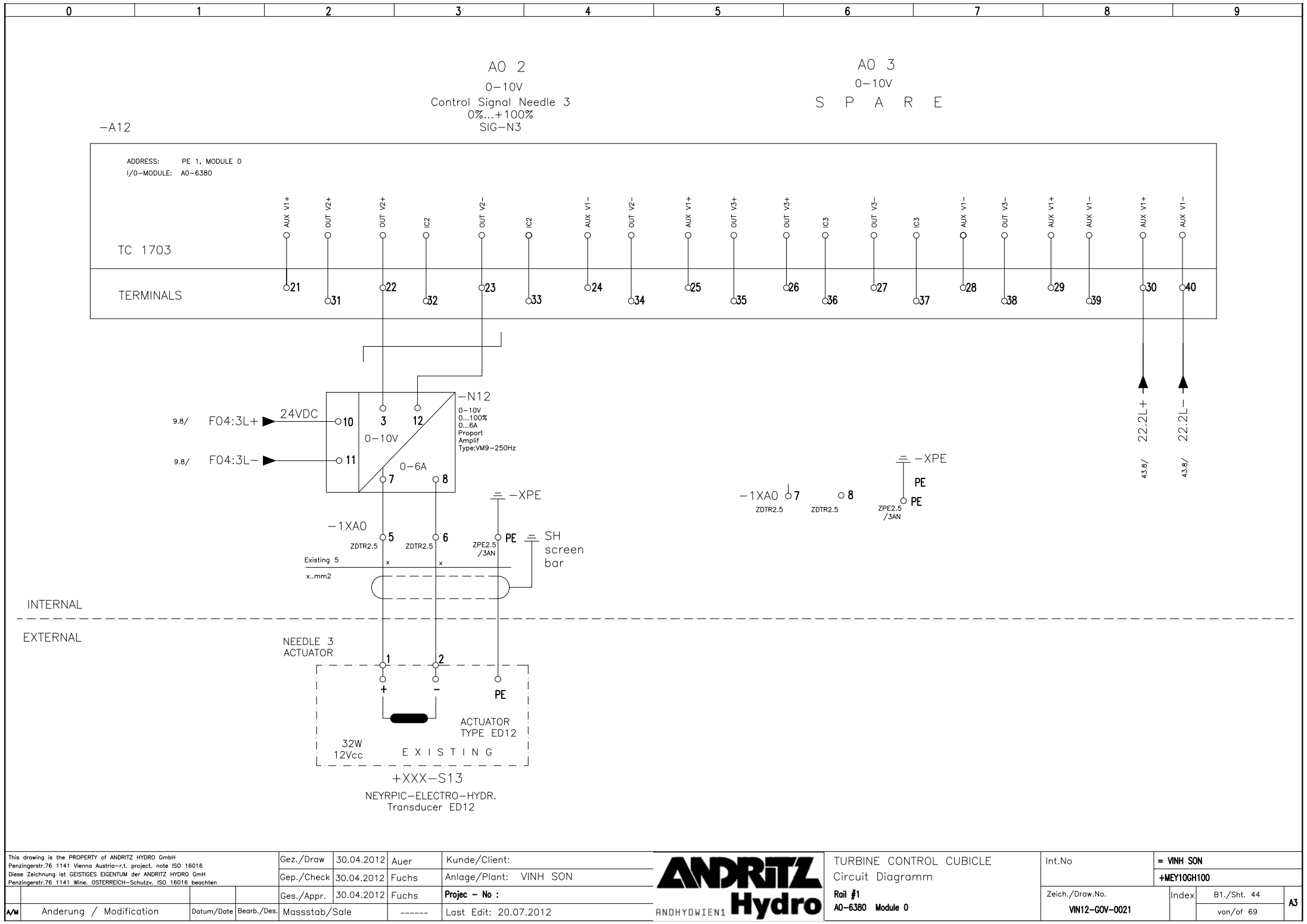
ADDRESS: PS 1, PS–MODULE
I/O–MODULE: PS–6630

TC 1703

TERMINALS







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Penzingerstr.76 1141 WIne. OSTERREICH-Schutzv. ISO 16016 beachten

Gez./Draw

30.04.2012

Auer

Kunde/Client:

Gep./Check

30.04.2012

Fuchs

Anlage/Plant: VINH SON

Gez./Appr.

30.04.2012

Fuchs

Projec - No :

Masstab/Sale

Last Edit: 20.07.2012

ANDRITZ

Hydro

ANDHYDWIEN1

TURBINE CONTROL CUBICLE

Circuit Diagramm

Rail #1

A0-6380 Module 0

Int.No

= VINH SON

Zeich./Draw.No.

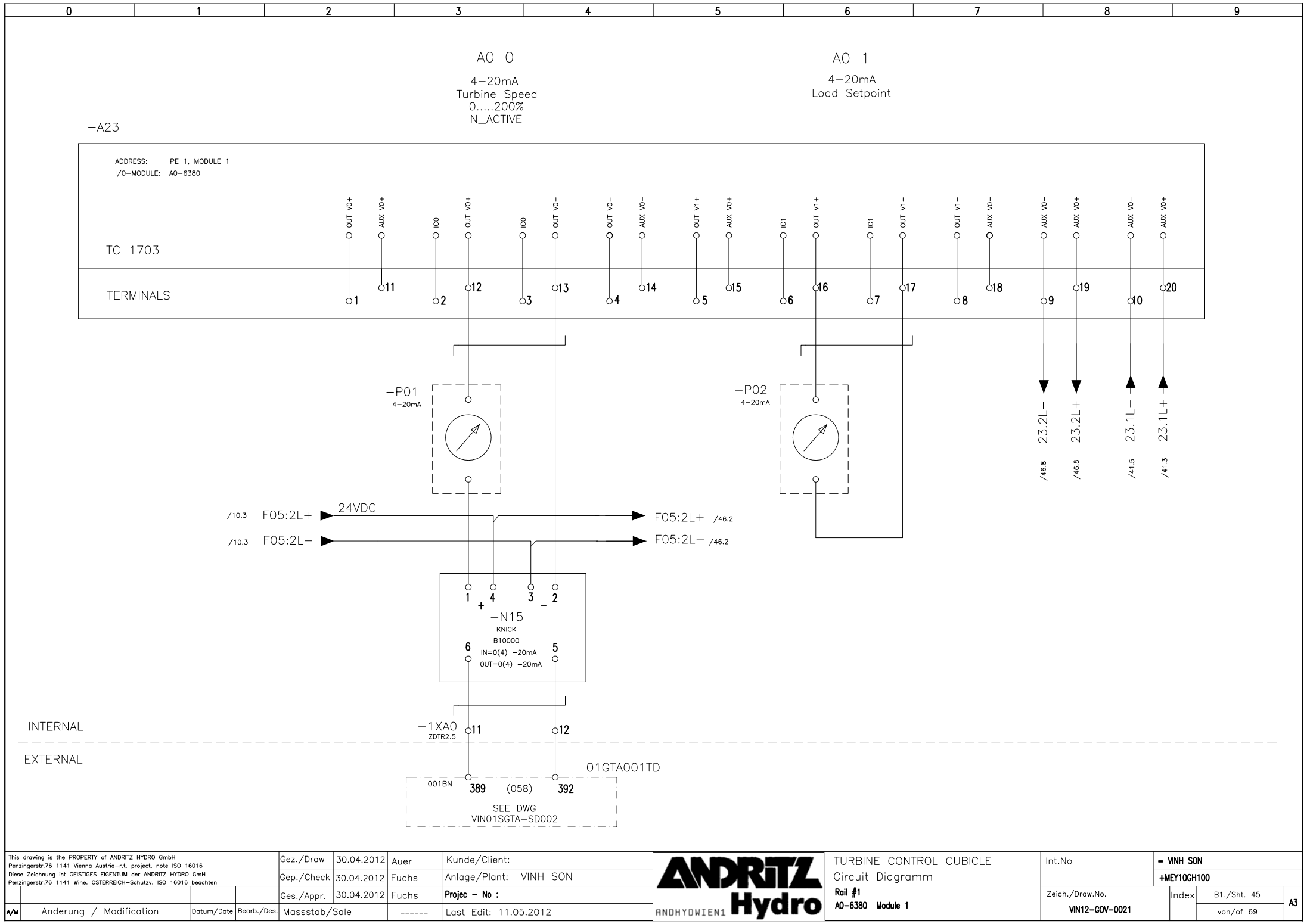
VIN12-G0V-0021

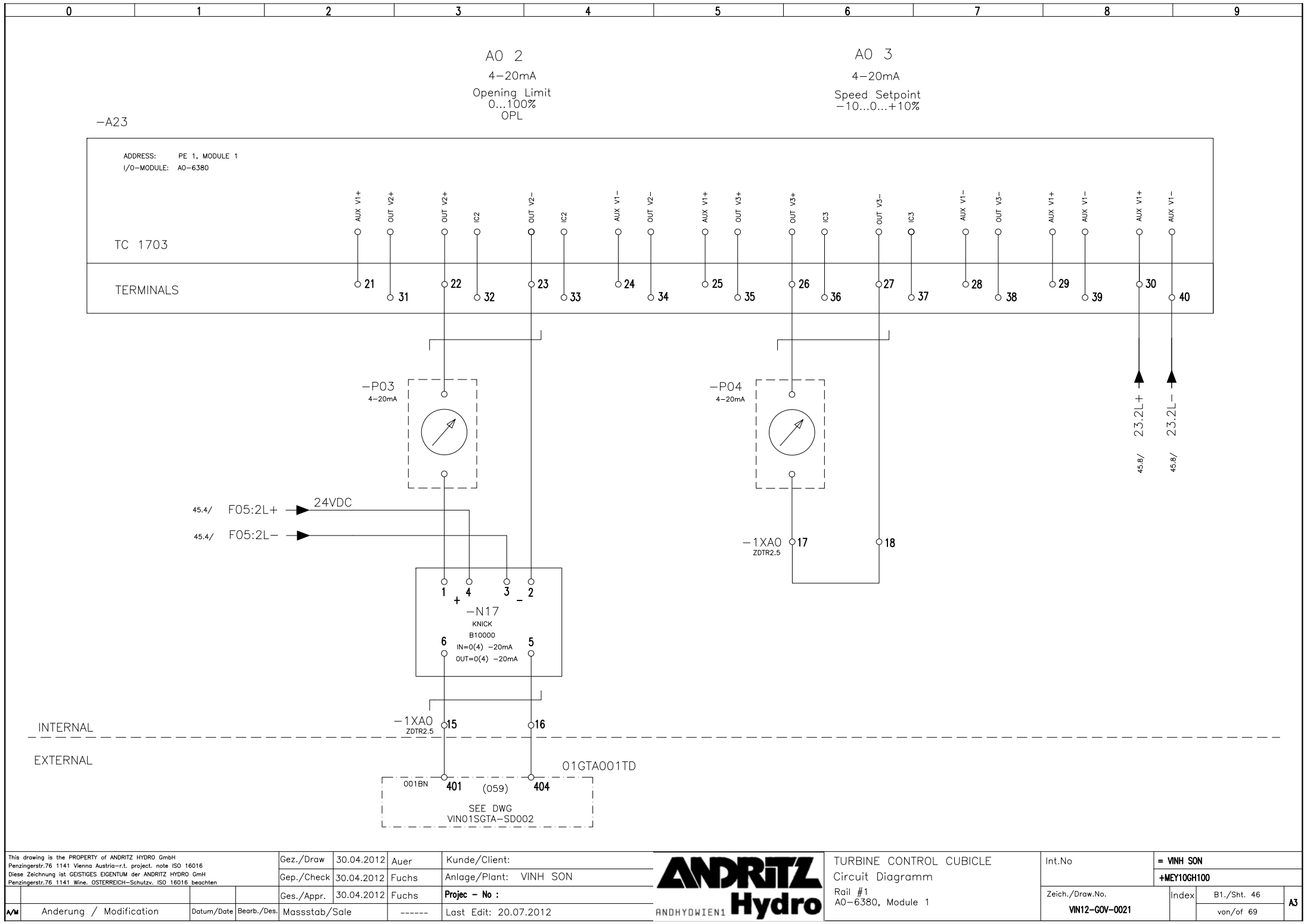
Index

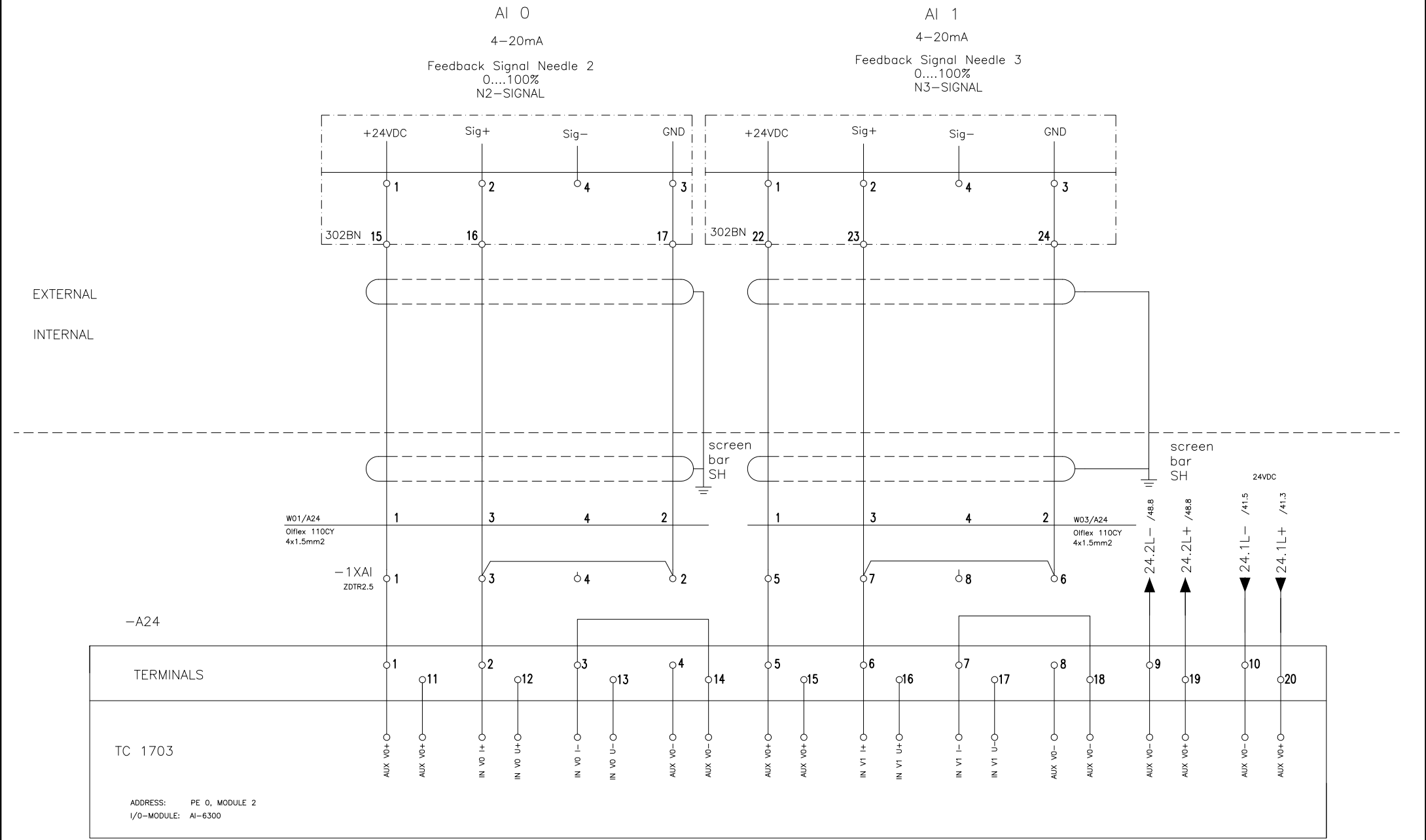
B1./Sht. 44

von/of 69

A3







A0 3
4-20mA
SPARE

INTERNAL



SMS 0

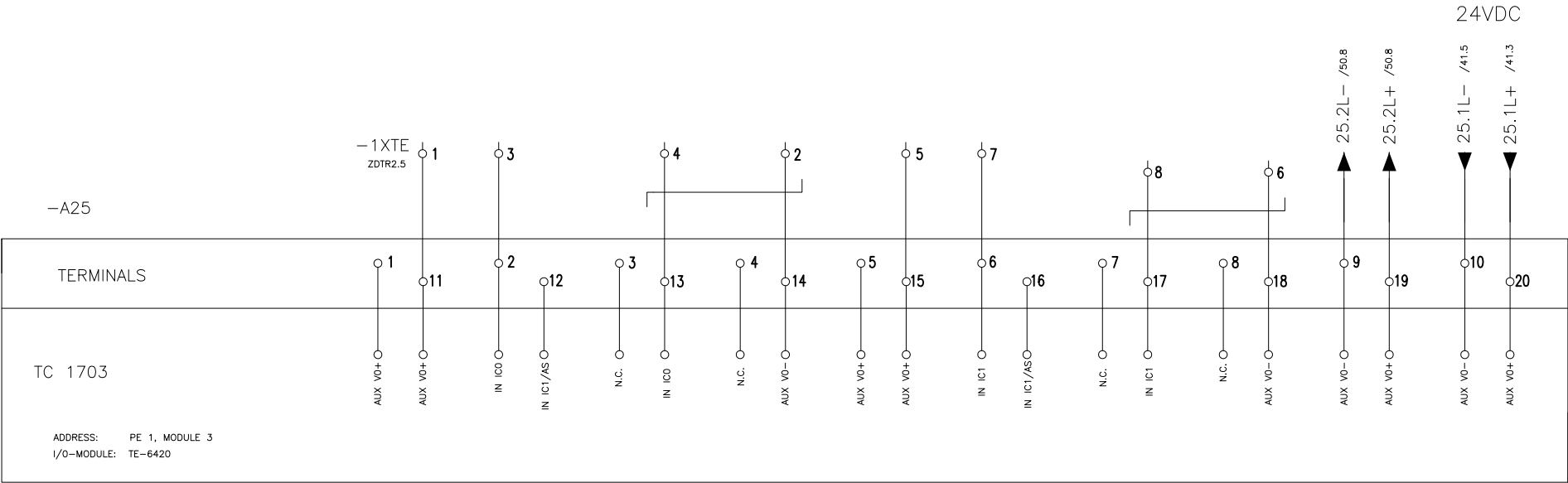
SMS 1

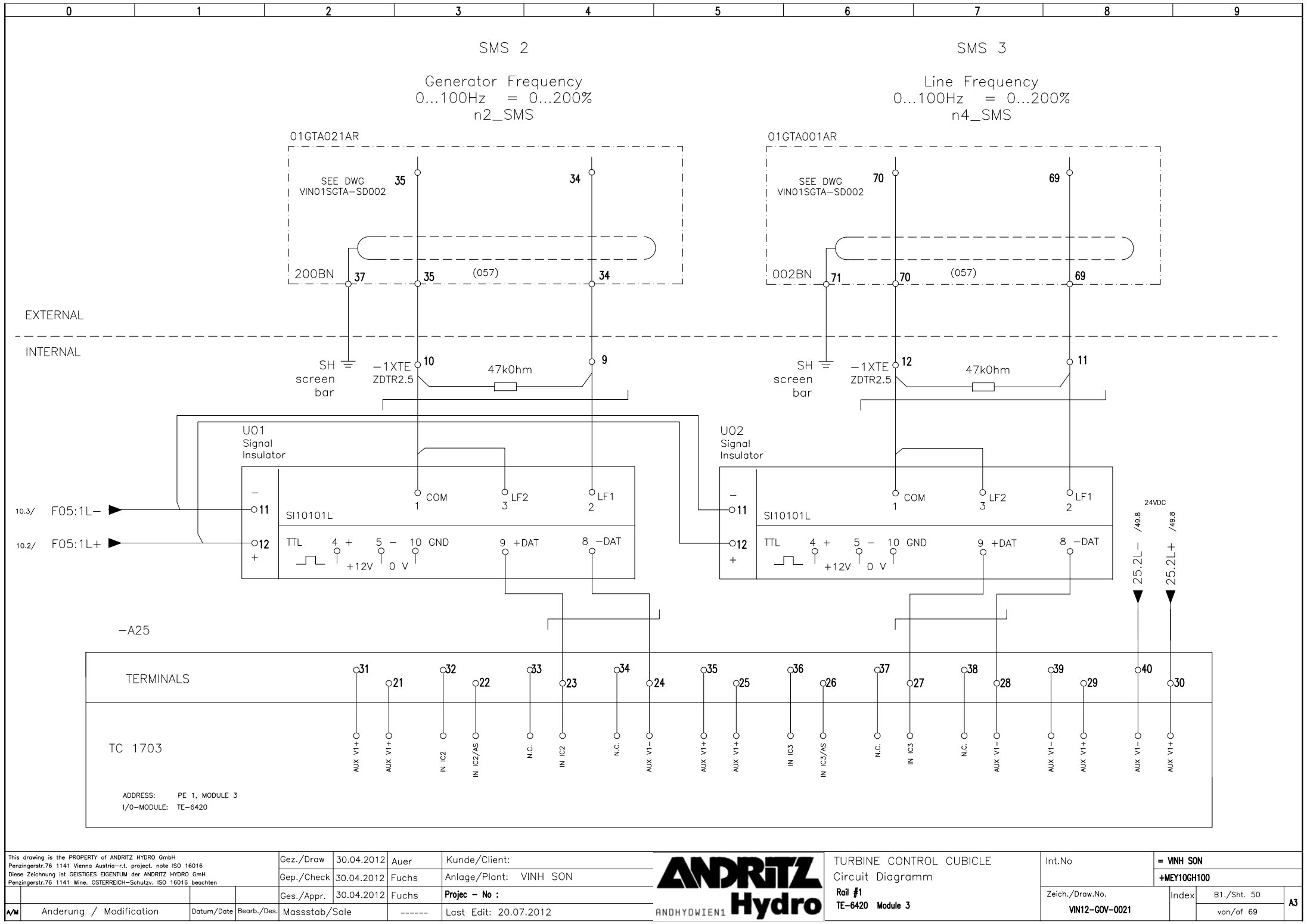
SPARE

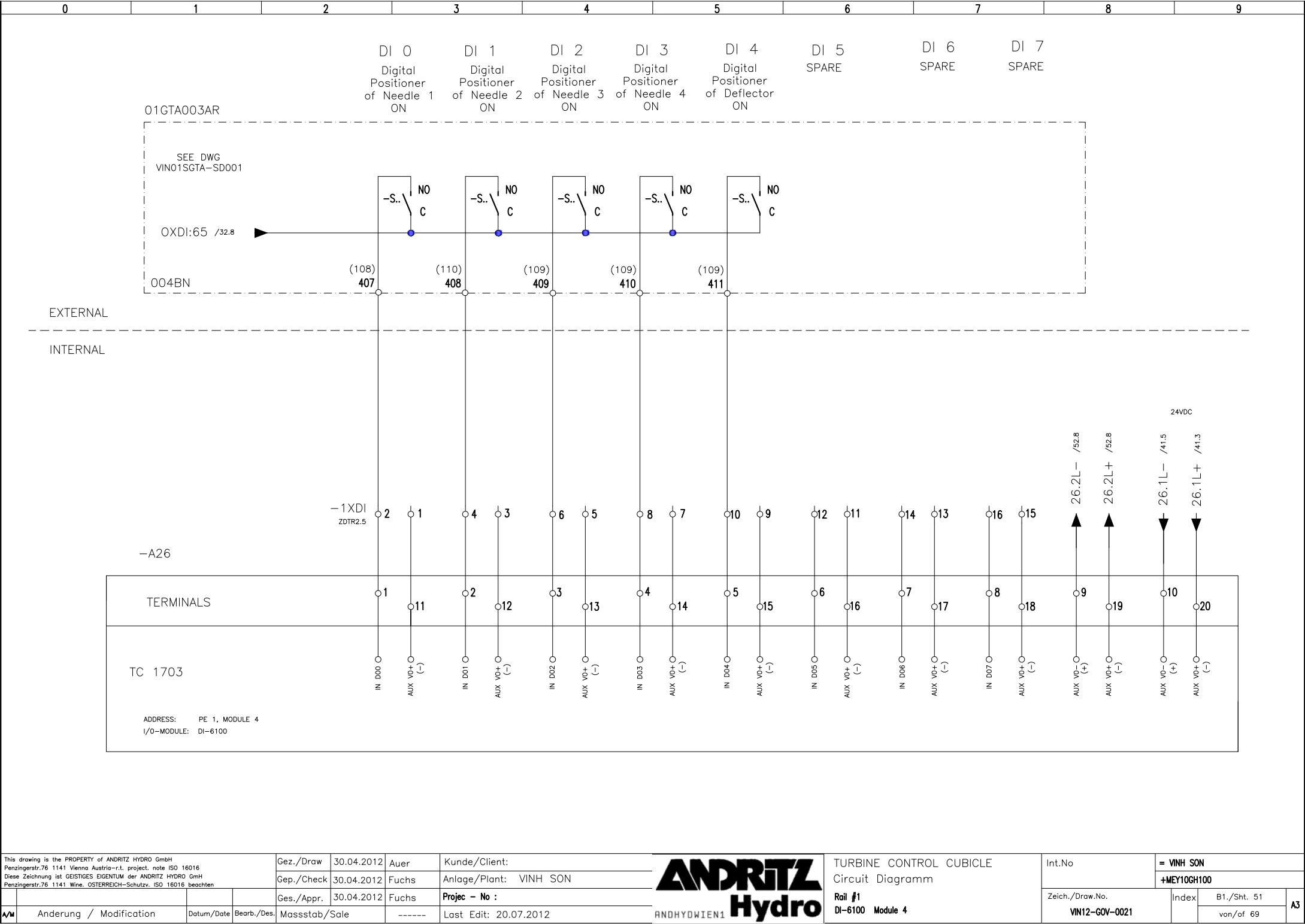
SPARE

EXTERNAL

INTERNAL

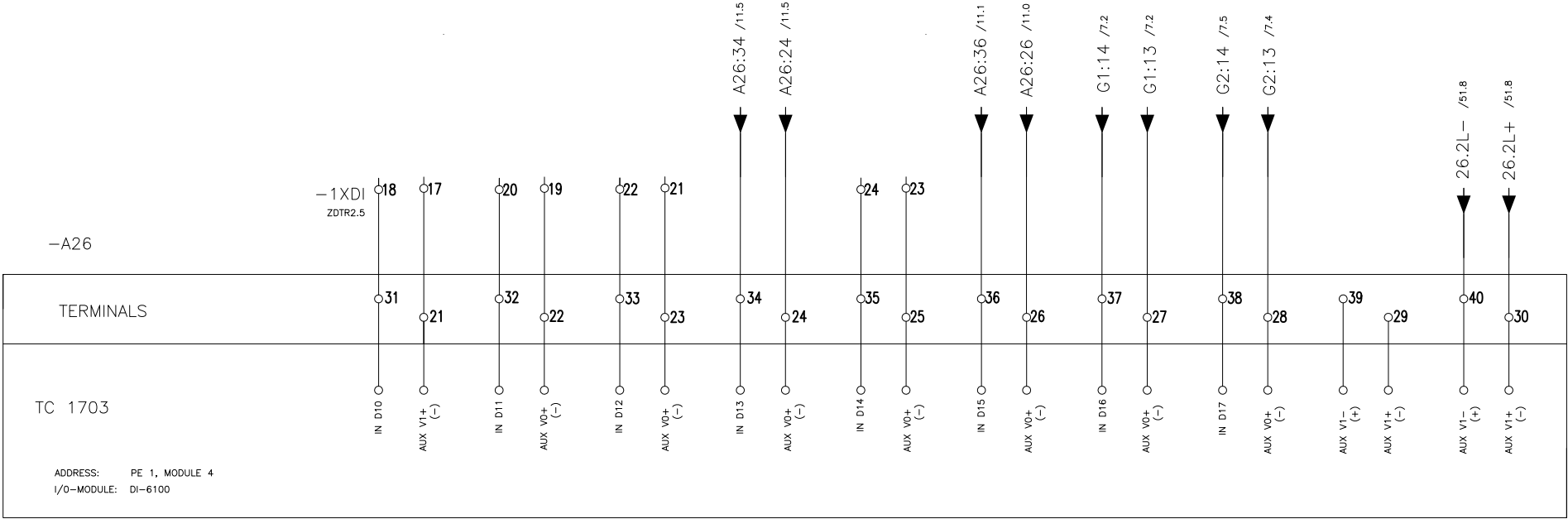






0	1	2	3	4	5	6	7	8	9
		DI 10 SPARE	DI 11 SPARE	DI 12 SPARE	DI 13 Load Circuit Controls Fault	DI 14 SPARE	DI 15 MCB Tripped F-FUSE	DI 16 Power Supply 1 Fault F-FUSE	DI 17 Power Supply 2 Fault F-PS2
EXTERNAL									

INTERNAL



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Penzingerstr.76 1141 WIne. ÖSTERREICH-Schutzv. ISO 16016 beachten

A/M		Datum/Date		Bearb./Des.		Masstab/Sale		Last Edit: 20.07.2012	
Änderung / Modification									

Gez./Draw 30.04.2012 Auer Kunde/Client:
Gep./Check 30.04.2012 Fuchs Anlage/Plant: VINH SON
Ges./Appr. 30.04.2012 Fuchs Projec - No :
Last Edit: 20.07.2012

ANDRITZ
Hydro
ANDHYDOWIEN1

TURBINE CONTROL CUBICLE
Circuit Diagramm
Rail #1
D0-6200 Module 5

Int.No		= VINH SON	
Zeich./Draw.No.		+MEY10GH100	
Index		B1./Sht. 53	
VIN12-GOV-0021		von/of 69	

INTERNAL

EXTERNAL

TC 1703

TERMINALS

DO 0 SPARE

DO 1 Needle 1 Positioner Faulty

DO 2 Needle 2 Positioner Faulty

DO 3 Needle 3 Positioner Faulty

DO 4 Needle 4 Positioner Faulty

-A27

ADDRESS: PE 1, MODULE 5
I/O-MODULE: D0-6200

OUT D00 AUX V0-

OUT D01 AUX V0-

OUT D02 AUX V0-

OUT D03 AUX V0-

OUT D04 AUX V0-

11 1

12 2

13 3

14 4

15 5

-K40 24VDC

-K41 24VDC

-K42 24VDC

-K43 24VDC

-K44 24VDC

A1 A2

Schrack PT570L24

14 11 67.1

12 21

24 21

22 31

34 31

32 41

44 41

42 41

Schrack PT570L24

14 11 67.3

12 21

24 21

22 31

34 31

32 41

44 41

42 41

Schrack PT570L24

14 11 67.4

12 21

24 21

22 31

34 31

32 41

44 41

42 41

Schrack PT570L24

14 11 67.1

12 21

24 21

22 31

34 31

32 41

44 41

42 41

Schrack PT570L24

14 11 67.3

12 21

24 21

22 31

34 31

32 41

44 41

42 41

Gez./Draw	30.04.2012	Auer	Kunde/Client:
Gep./Check	30.04.2012	Fuchs	Anlage/Plant: VINH SON
Ges./Appr.	30.04.2012	Fuchs	Projec - No :
Massstab/Sale		-----	Last Edit: 20.07.2012

ANDRITZ
ANDHYDWIEN1 **Hydro**

TURBINE CONTROL CUBICLE
Circuit Diagram
Rail #1
D0-6200 Module 5

Int.No		= VINH SON		A3
		+MEY10GH100		
Zeich./Draw.No. VIN12-GOV-0021		Index	B1./Sht. 53 von/of 69	A3

-A27

D0 5
Deflector
Positioner
Faulty

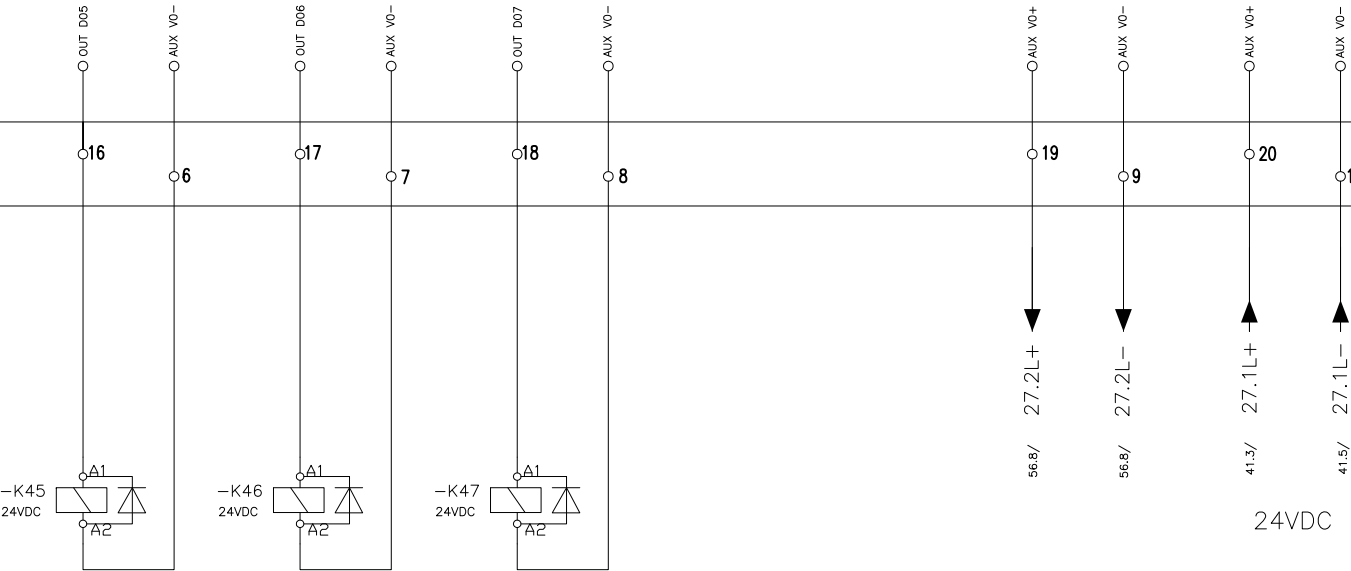
D0 6
Circuit
Braker
Closed

D0 7
SPARE

ADDRESS: PE 1, MODULE 5
I/O-MODULE: AO-6200

TC 1703

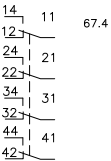
TERMINALS



INTERNAL

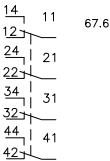
EXTERNAL

Schrack
PT570L24



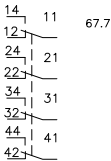
67.4

Schrack
PT570L24

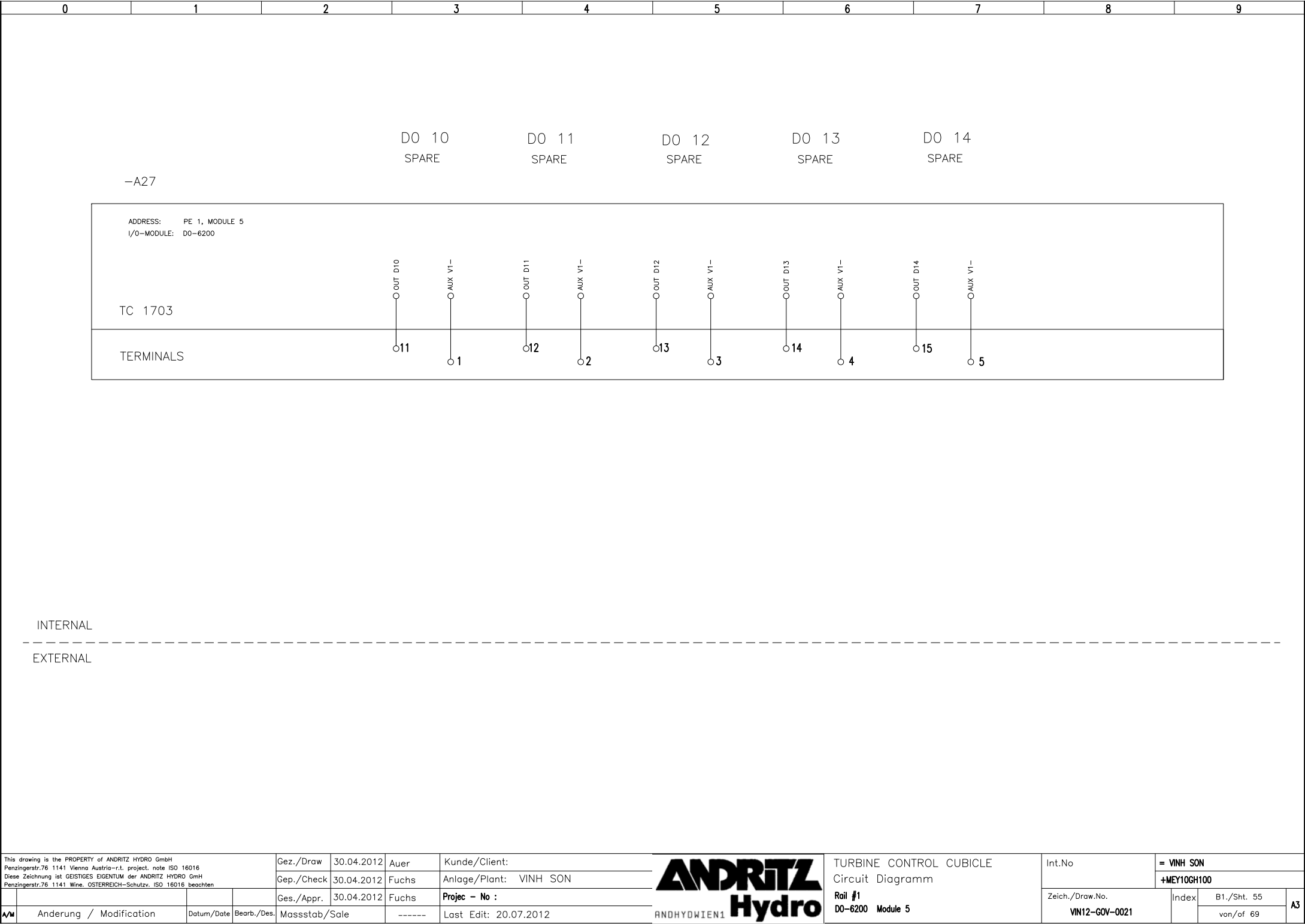


67.6

Schrack
PT570L24



67.7



INTERNAL

EXTERNAL

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Gez./Draw 30.04.2012 Auer

Gep./Check 30.04.2012 Fuchs

Ges./Appr. 30.04.2012 Fuchs

Kunde/Client:

Anlage/Plant: VINH SON

Projec - No :

Last Edit: 20.07.2012

ANDRITZ
Hydro

ANDHYDWIEN1

TURBINE CONTROL CUBICLE
Circuit Diagramm

Rail #1
DO-6200 Module 5

Int.No

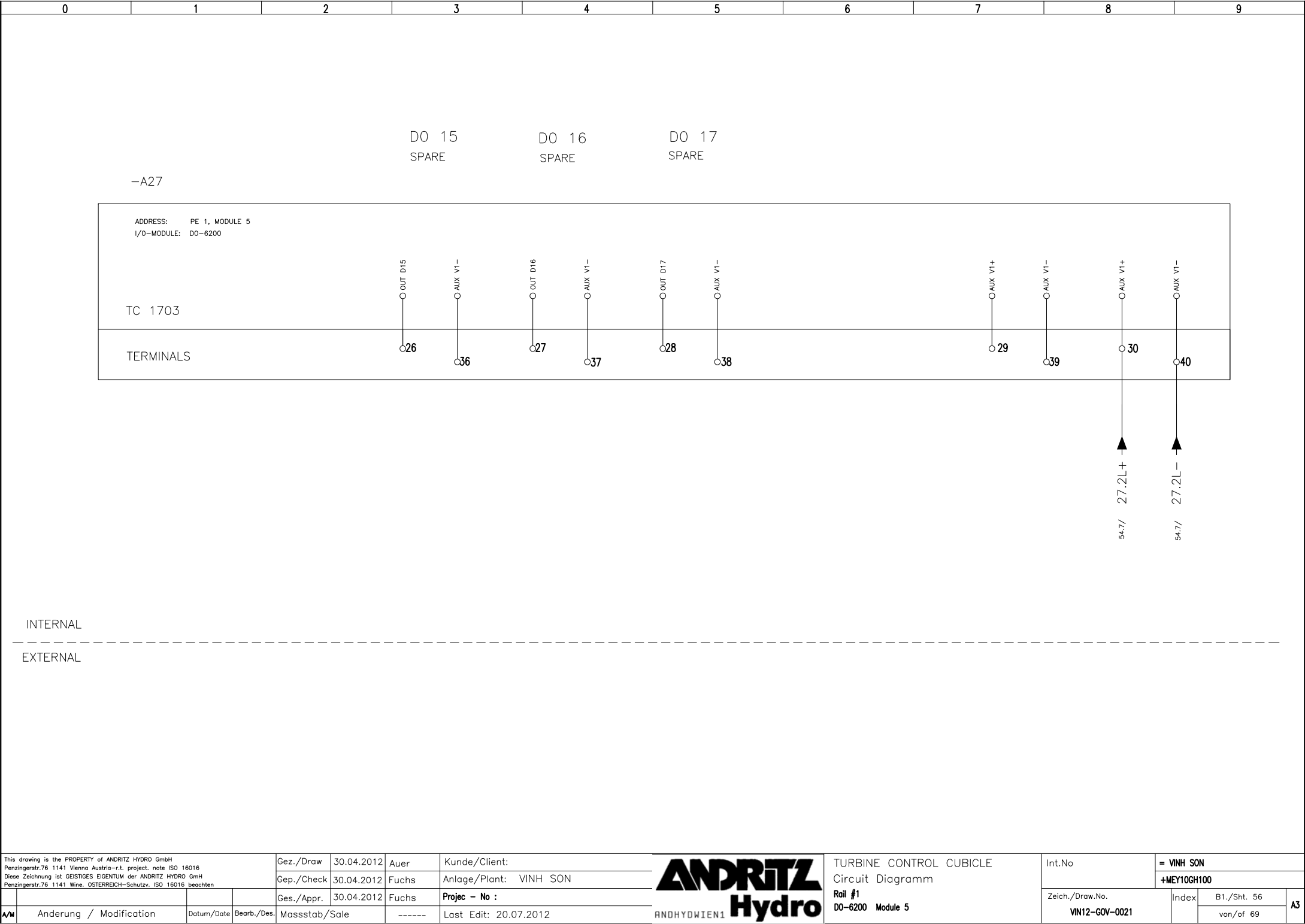
Zeich./Draw.No.

= VINH SON
+MEY10GH100

Index

B1./Sht. 55
von/of 69

A3

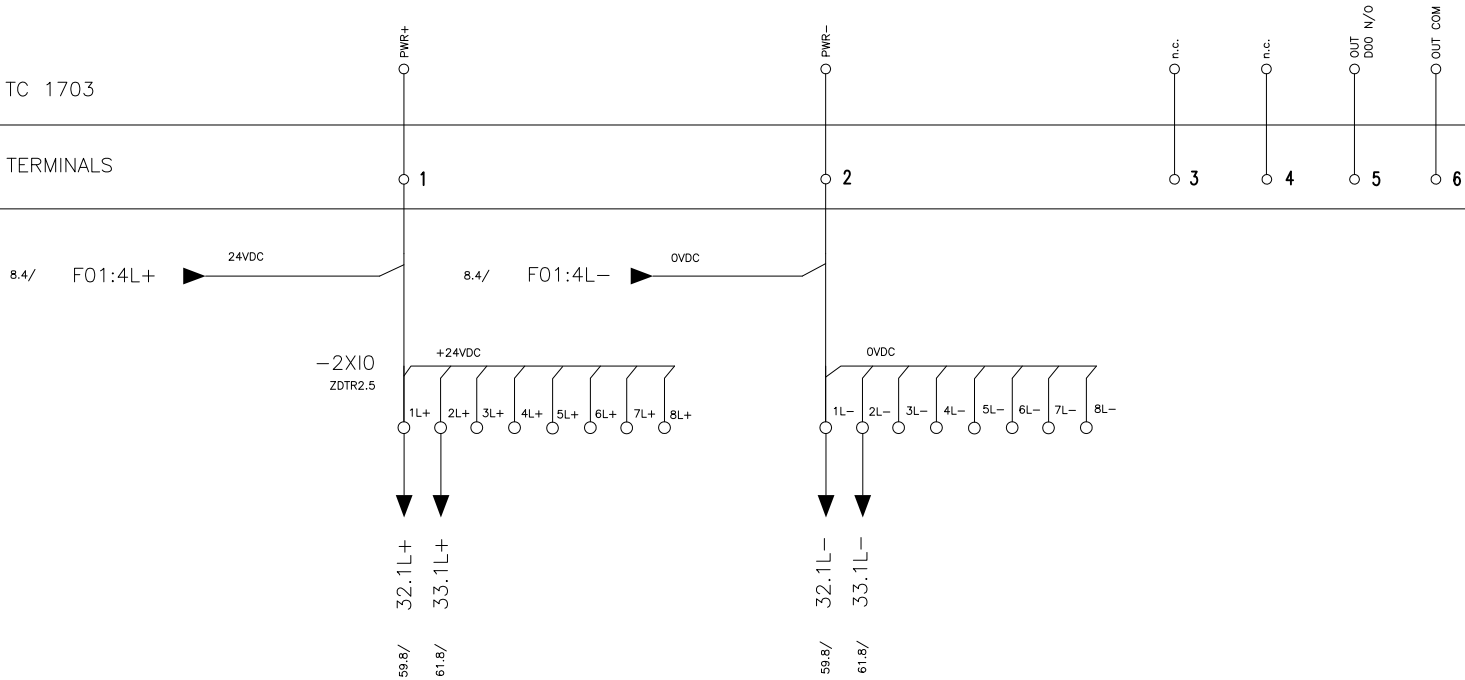


–A30

ADDRESS: PS 2, PS-MODULE
I/O-MODULE: PS-6630

TC 1703

TERMINALS



AO 0
4-20mA
Position of Needle 1
0...100%
N1_INT

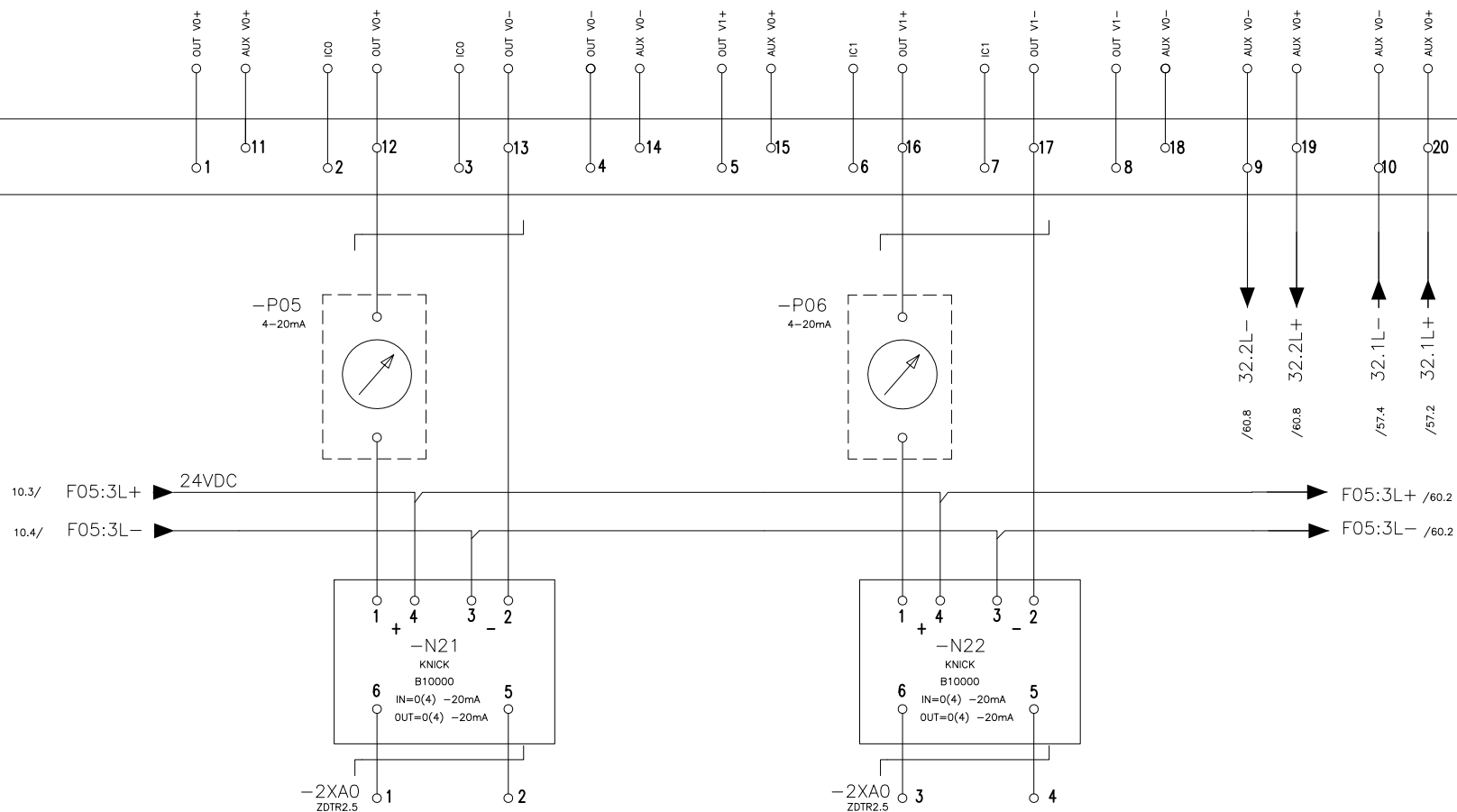
AO 1
4-20mA
Position of Needle 2
0...100%
N2_INT

—A32

ADDRESS: PE 2, MODULE 0
I/O-MODULE: A0-6380

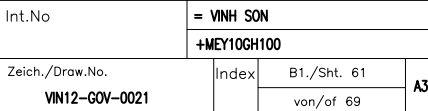
TC 1703

TERMINALS



INTERNAL

EXTERNAL



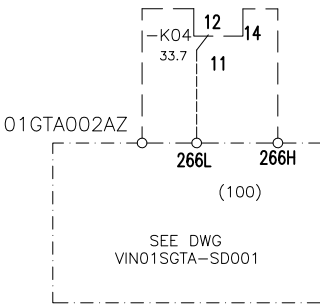
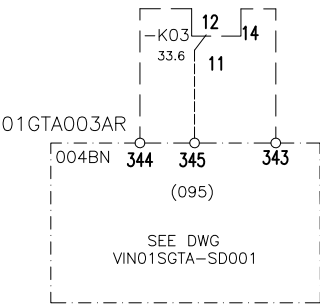
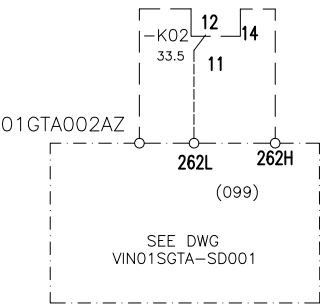
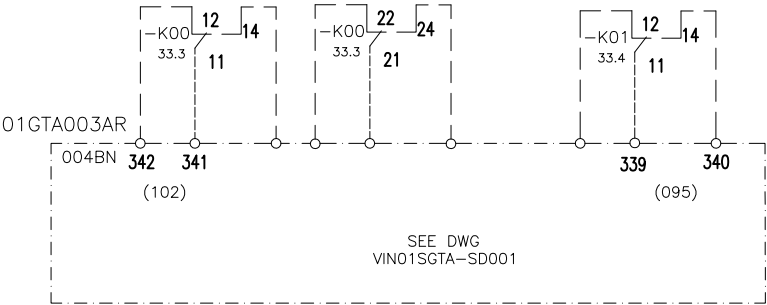
Major Fault
No Turbine Governor
Shut Down

MInor Fault
General
Alarm

Turbine
Limited

Isolating
Operation
Active

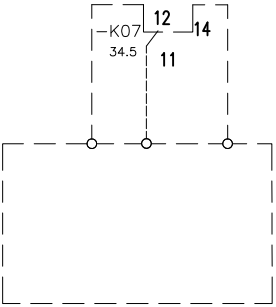
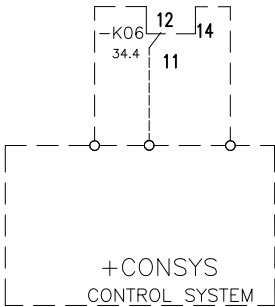
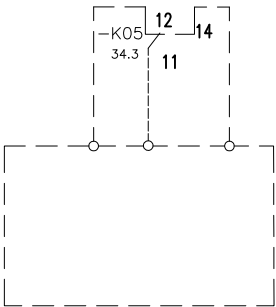
Speed
Controller
Active

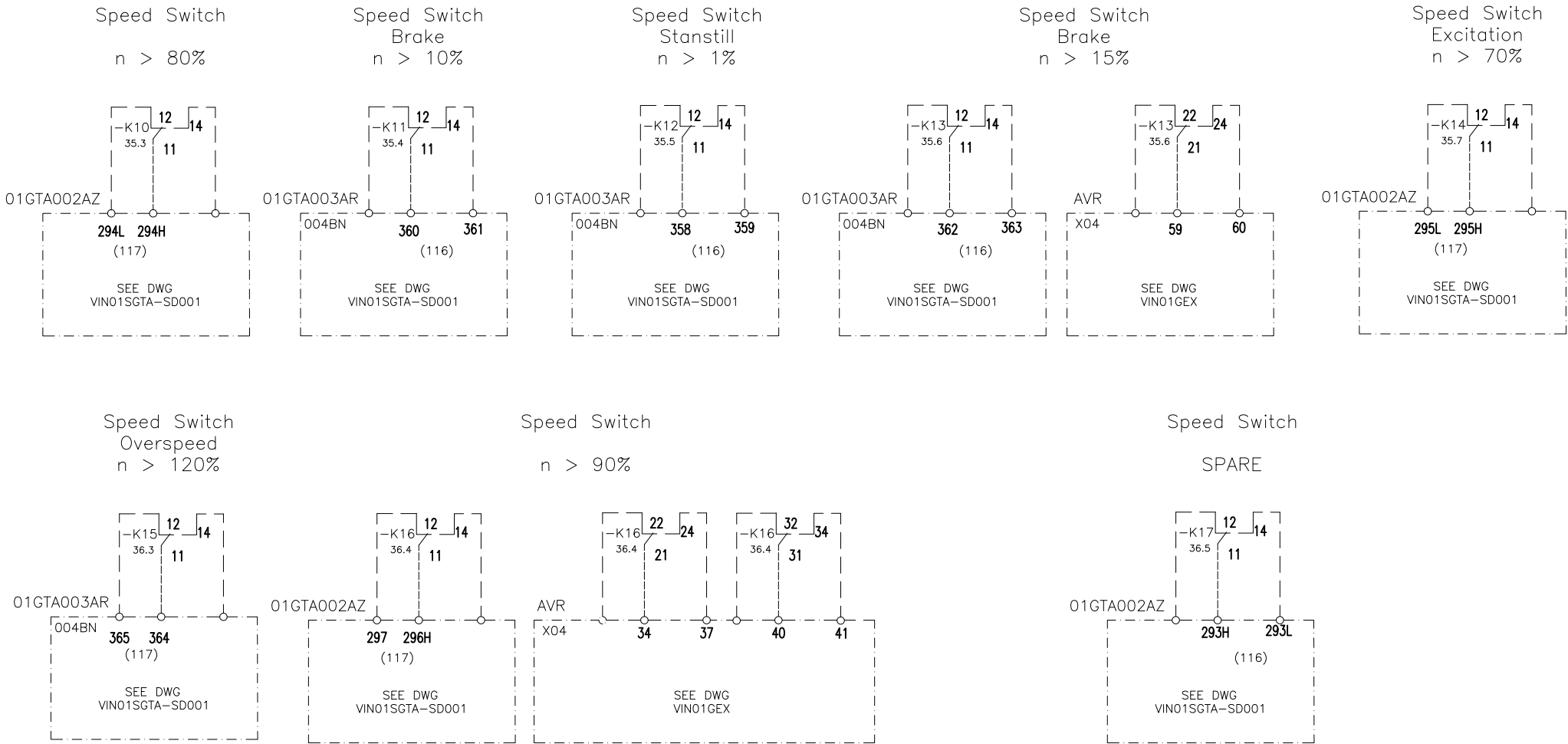


Flow Controller
Active

Power Controller
Active

TC Ready
To Start





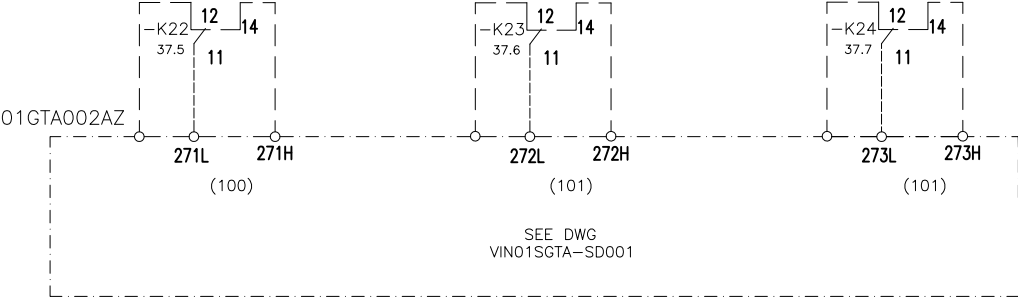
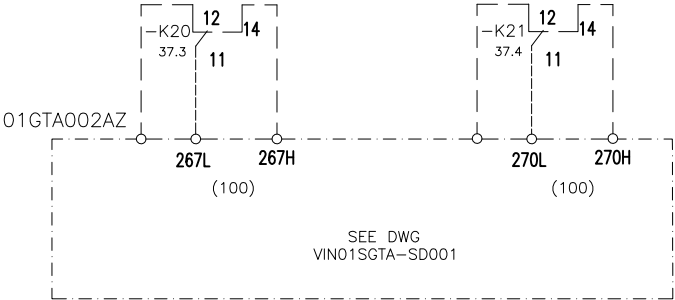
Needle 1
not bloked

Needle 2
not bloked

Needle 3
not bloked

Needle 4
not bloked

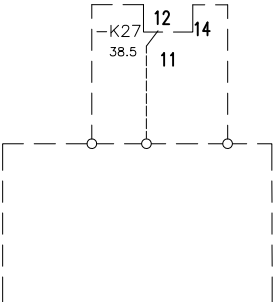
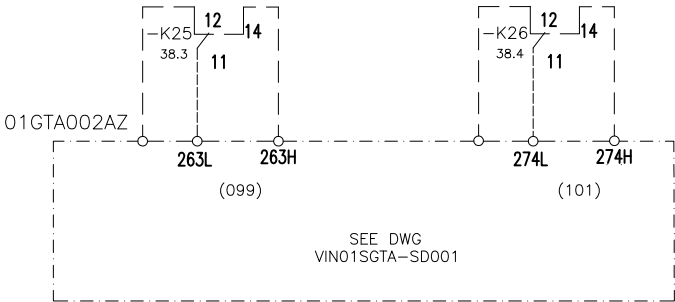
Deflector
not bloked



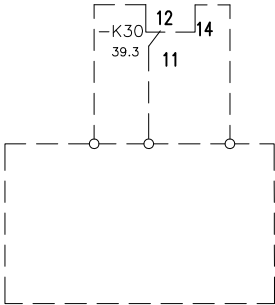
Opening Limited
at Max. Position

Power Less
Than 1MW

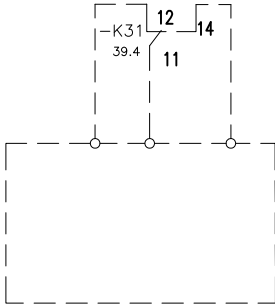
Governor in
Automatic Mode



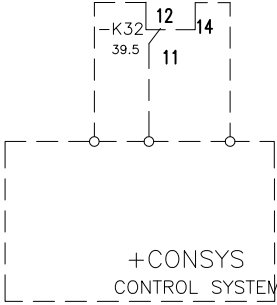
Ready for Load
Take Up



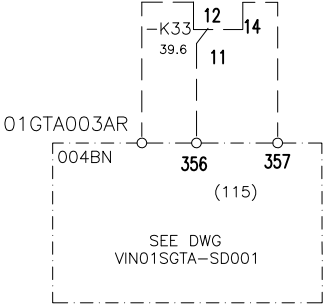
Unit in
Operation



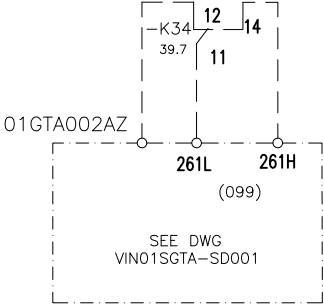
Black Start
Active



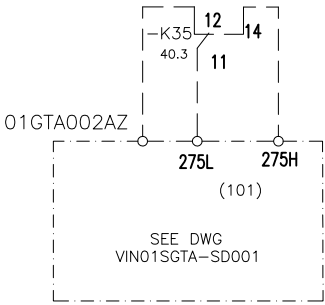
Tachometer
fault



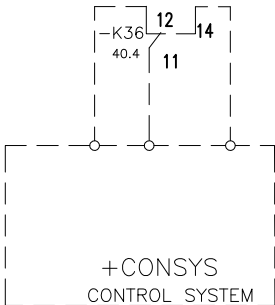
Speed Noload
position



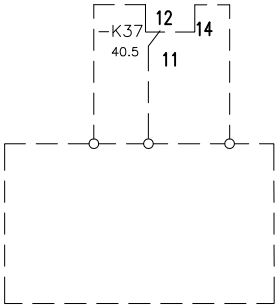
Start LO_ON
feedback



Primary Control
Active



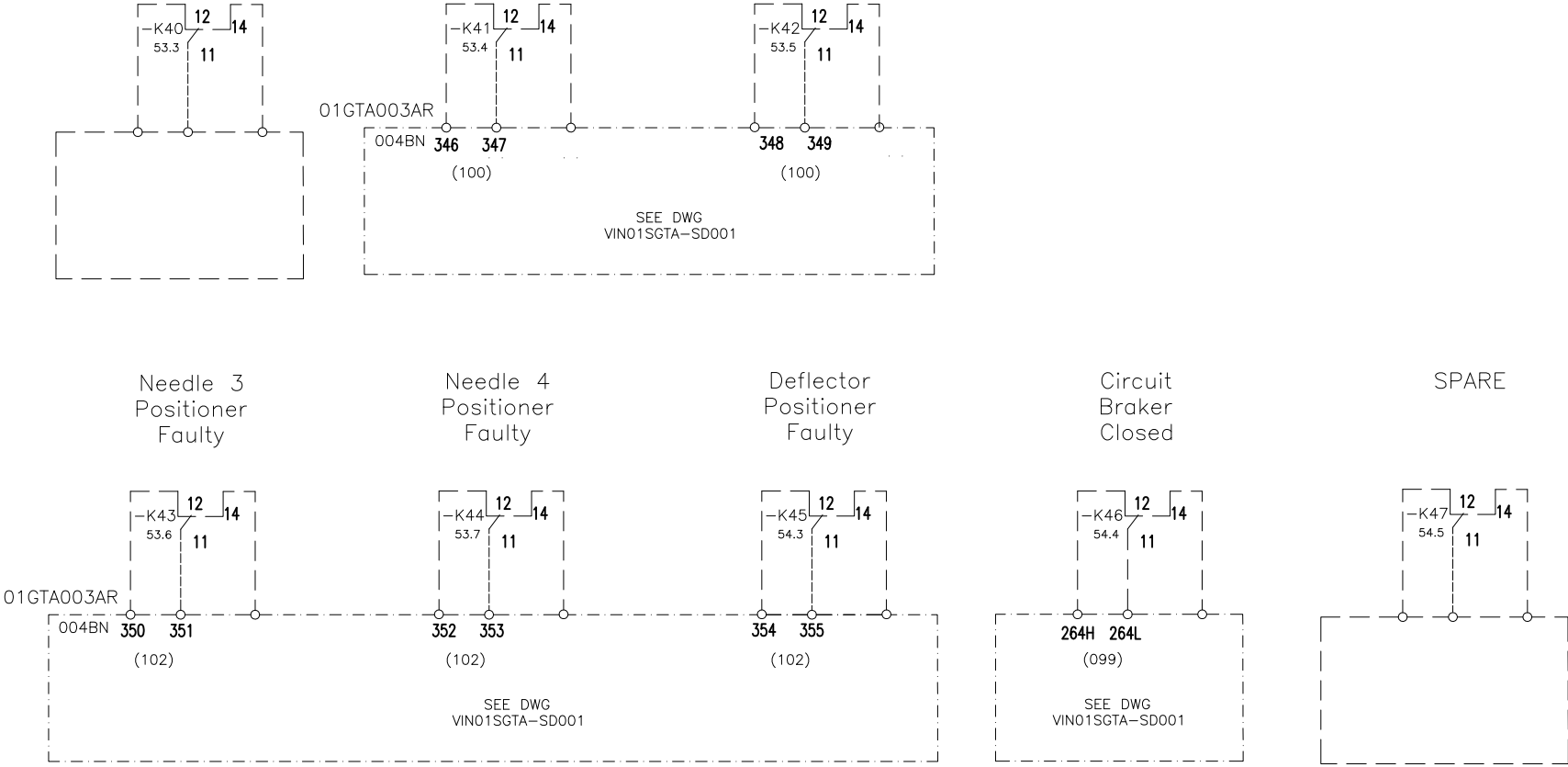
Nozzles
Are Closed



SPARE

Needle 1
Positioner
Faulty

Needle 2
Positioner
Faulty



Needle 3
Positioner
Faulty

Needle 4
Positioner
Faulty

Deflector
Positioner
Faulty

Circuit
Braker
Closed

SPARE

